MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE, NEGATIVE, VOLTAGE REGULATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for three- and four-terminal, monolithic silicon, adjustable, negative, voltage regulators. Three product assurance classes and a choice of lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Case outline letter
01	4-terminal voltage regulator, -30 volts $\leq V_O \leq -5$ volts at 0.5 A	X
02	4-terminal voltage regulator, -30 volts $\leq V_O \leq -5$ volts at 1.0 A	Y
03	3-terminal voltage regulator, -37 volts $\leq V_O \leq -1.25$ volts at 0.5 A	X
04	3-terminal voltage regulator, -37 volts $\leq V_O \leq -1.25$ volts at 1.5 A	Y

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline
X (device type 01)	See figure 1
Y (device type 02)	See figure 2
X (device type 03)	See figure 3
Y (device type 04)	See figure 4

1.3 Absolute maximum ratings.

Input voltage (device types 01 and 02)	-40 Vdc
Input-output differential voltage (device types 03 and 04)	-40 Vdc
Lead temperature (soldering, 60 seconds)	300°C
Junction temperature	T_J = 150°C 1/
Storage temperature	-65°C to +150°C

1.4 Recommended operating conditions.

Input voltage range:	
Device types 01 and 02	-8 to -38 Vdc
Device types 03 and 04	-4.25 to -41.25 Vdc
Ambient operating temperature range	-55°C to +125°C

1/ The device is protected by a thermal shutdown circuit which is designed to turn off the output transistor whenever the device junction temperature is in excess of 150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Power and thermal characteristics.

T_A = T_S	Case	Max θ_{J-A}	Maximum P_D without heat sink	Max θ_{J-C}	Maximum P_D with heat sink	Max θ_{C-S} 2/
125°C 3/	X	140°C/W	0.18 W	40°C/W	0.5 W	10°C/W
	Y	35°C/W	0.71 W	4°C/W	5.6 W	0.5°C/W
25°C	X	140°C/W	0.89 W	40°C/W	2.50 W	10°C/W
	Y	35°C/W	3.60 W	4°C/W	28.00 W	0.5°C/W
-55°C	X	140°C/W	1.50 W	40°C/W	4.00 W	10°C/W
	Y	35°C/W	5.80 W	4°C/W	45.00 W	0.5°C/W

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Block diagrams and terminal connections. The block diagrams and terminal connections shall be as specified on figures 5 through 8.

3.2.2 Schematic circuits. The schematic circuits shall be as specified on figure 9.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3 and on figures 1 through 4.

3.3 Lead material and lead finish.

2/ This value represents the maximum allowable thermal impedance of a heat sink to remain within the above thermal ratings.

3/ Based on T_J = 150°C and specified values of θ_{J-A} and θ_{J-C}.

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TABLE I. Electrical performance characteristics for device type 01 (see 3.4 unless otherwise specified). 1/ 2/

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = -8V	I_L = 5mA, 500mA		-5.25	-4.75	V
		V_{IN} = -30V	I_L = 5mA, 50mA		-5.25	-4.75	V
		V_{IN} = -38V	I_L = 500mA		-31.5	-28.5	V
		V_{IN} = -10V	I_L = 5mA	T_A = 150°C	-5.25	-4.75	V
Line regulation	V_{RLINE}	-30V≤V_{IN}≤-8V	I_L = 50mA		-150	150	mV
		-25V≤V_{IN}≤-8V	I_L = 350mA		-50	50	mV
Load regulation	V_{RLOAD}	V_{IN} = -10V	5mA ≤ I_L ≤ 500mA		-100	100	mV
		V_{IN} = -30V	5mA ≤ I_L ≤ 50mA		-150	150	mV
Thermal regulation	V_{RTH}	V_{IN} = -15V	I_L = 500mA	T_A = 25°C	-50	50	mV
Standby current drain	I_{SCD}	V_{IN} = -10V	I_L = 5mA		0.1	3.0	mA
		V_{IN} = -30V	I_L = 5mA		0.1	4.0	mA
Standby current drain change versus line voltage	ΔI_{SCD} (LINE)	-30V≤V_{IN}≤-8V	I_L = 5mA		-1.0	1.0	mA
Standby current drain change versus load current	ΔI_{SCD} (LOAD)	V_{IN} = -10V	5mA≤ I_L≤500mA		-0.5	0.5	mA
Control pin current	I_{CTL}	V_{IN} = -10V	I_L = 350mA	T_A = 25°C	0.001	2.00	µA
				-55°C ≤ T_A ≤ 125°C	0.001	3.00	µA
Output short circuit current	I_{OS1}	V_{IN} = -10V			0.002	2.0	A
	I_{OS2}	V_{IN} = -30V			0.002	1.0	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = -10V	R_L = 10Ω, C_L = 20 µF	after I_{OS1} 3/	-5.25	-4.75	V
		V_{IN} = -30V	R_L = 5 kΩ	after I_{OS2} 3/	-5.25	-4.75	V
Voltage start-up	V_{START}	V_{IN} = -20V	R_L = 10Ω, C_L = 20µF		-5.25	-4.75	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -10V e_i = 1 Vrms @ f = 2400 Hz	I_L = 125mA	T_A = 25°C 4/ (figure 13)	45	-	dB
Output noise voltage	V_{NO}	V_{IN} = -10V	I_L = 50mA	T_A = 25°C 4/ (figure 14)	-	250	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -10V V_{pulse} = -3.0V	I_L = 5mA	T_A = 25°C 5/ (figure 15)	-	30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = -10V	I_L = 50mA ΔI_L = 200mA	T_A = 25°C 5/ (figure 16)	-	2.5	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = -8V	I_L = 5mA, 1A		-5.25	-4.75	V
		V_{IN} = -30V	I_L = 5mA, 100mA		-5.25	-4.75	V
		V_{IN} = -38V	I_L = 1A		-31.5	-28.5	V
		V_{IN} = -10V	I_L = 5mA	T_A = 150°C	-5.25	-4.75	V
Line regulation	V_{RLINE}	-30V ≤ V_{IN} ≤ -8V	I_L = 100mA		-150	150	mV
		-25V ≤ V_{IN} ≤ -8V	I_L = 500mA		-75	75	mV
Load regulation	V_{RLOAD}	V_{IN} = -10V	5mA ≤ I_L ≤ 1A		-100	100	mV
		V_{IN} = -30V	5mA ≤ I_L ≤ 100mA		-150	150	mV
Thermal regulation	V_{RTH}	V_{IN} = -15V	I_L = 1A	T_A = 25°C	-50	50	mV
Standby current drain	I_{SCD}	V_{IN} = -10V	I_L = 5mA		0.1	3.0	mA
		V_{IN} = -30V	I_L = 5mA		0.1	4.0	mA
Standby current drain change versus line voltage	ΔI_{SCD} (LINE)	-30V ≤ V_{IN} ≤ -8V	I_L = 5mA		-1.0	1.0	mA
Standby current drain change versus load current	ΔI_{SCD} (LOAD)	V_{IN} = -10V	5mA < I_L < 1A		-0.5	0.5	mA
Control pin current	I_{CTL}	V_{IN} = -10V	I_L = 500mA	T_A = 25°C	0.01	2.00	µA
				-55°C ≤ T_A ≤ 125°C	0.001	3.00	µA
Output short circuit current	I_{OS1}	V_{IN} = -10V			0.002	4.5	A
	I_{OS2}	V_{IN} = -30V			0.002	2.0	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = -10V	R_L = 5Ω, C_L = 20 µF	after I_{OS1} 3/	-5.25	-4.75	V
		V_{IN} = -30V	R_L = 5 kΩ	after I_{OS2} 3/	-5.25	-4.75	V
Voltage start-up	V_{START}	V_{IN} = -20V	R_L = 5Ω, C_L = 20µF		-5.25	-4.75	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -10V, e_i = 1 Vrms @ f = 2400 Hz	I_L = 350mA	T_A = 25°C 4/ (figure 13)	45	-	dB
Output noise voltage	V_{NO}	V_{IN} = -10V	I_L = 100mA	T_A = 25°C 4/ (figure 14)	-	250	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -10V, V_{pulse} = -3.0V	I_L = 5mA	T_A = 25°C 5/ (figure 15)	-	30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = -10V	I_L = 100mA, ΔI_L = 400mA	T_A = 25°C 5/ (figure 16)	-	2.5	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = -4.25V	I_L = 5mA, 500mA	T_A = 25°C	-1.275	-1.225	V
				-55°C ≤ T_A ≤125°C	-1.30	-1.20	V
		V_{IN} = -41.25V	I_L = 5mA, 50mA	T_A = 25°C	-1.275	-1.225	V
				-55°C ≤ T_A ≤125°C	-1.30	-1.20	V
		V_{IN} = -6.25V	I_L = 5mA	T_A = 150°C	-1.30	-1.20	V
Line regulation	V_{RLINE}	-41.25V ≤V_{IN} ≤-4.25V	I_L = 5mA	T_A = 25°C	-9	9	mV
				-55°C ≤T_A ≤125°C	-23	23	mV
Load regulation	V_{RLOAD}	V_{IN} = -6.25V	5mA ≤I_L ≤200mA	T_A = 25°C	-6	6	mV
				-55°C ≤T_A ≤125°C	-12	12	mV
		V_{IN} = -6.25V	5mA ≤I_L ≤500mA	T_A = 25°C	-12	12	mV
				-55°C ≤T_A ≤125°C	-24	24	mV
		V_{IN} = -41.25V	5mA ≤I_L ≤50mA	T_A = 25°C	-6	6	mV
				-55°C ≤T_A ≤125°C	-12	12	mV
Thermal regulation	V_{RTH}	V_{IN} = -14.6V	I_L = 750mA	T_A = 25°C	-5	5	mV
Adjust pin current	I_{ADJ}	V_{IN} = -4.25V	I_L = 5mA		25	100	µA
		V_{IN} = -41.25V	I_L = 5mA		25	100	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	-41.25V ≤V_{IN} ≤-4.25V	I_L = 5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = -6.25V	5mA ≤I_L≤ 500mA		-5	5	µA
Minimum load current	I_Q	-14.25V ≤V_{IN} ≤ -4.25V forced V_{OUT} = -1.4V			0.20	3.00	mA
		V_{IN} = -41.25V forced V_{OUT} = -1.4V			1.00	5.00	mA
Output short circuit current	I_{OS1}	V_{IN} = -4.25V			0.5	1.8	A
	I_{OS2}	V_{IN} = -40V			0.05	0.5	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = -4.25V	R_L = 2.5Ω, C_L = 20 µF	after I_{OS1} 3/	-1.30	-1.20	V
		V_{IN} = -40V	R_L = 250Ω	after I_{OS2} 3/	-1.30	-1.20	V
Voltage start-up	V_{START}	V_{IN} = -4.25V	R_L = 2.5Ω, C_L = 20µF		-1.30	-1.20	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -6.25V e_i = 1 Vrms @ fo =2400 Hz	I_L = 125mA	T_A = 25°C 4/ (figure 13)	48	-	dB
Output noise voltage	V_{NO}	V_{IN} = -6.25V	I_L = 50mA	T_A = 25°C 4/ (figure 14)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -6.25V V_{IN} = -1.0V	I_L = 10mA	T_A = 25°C 5/ (figure 15)	-	80	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = -6.25V	I_L = 50mA ΔI_L = 200mA	T_A = 25°C 5/ (figure 16)	-	0.30	mV/mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 04 (see 3.4 unless otherwise specified) 1/2/-Continued.

Characteristic	Symbol	Condition: (fig. 12 unless otherwise stated)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = -4.25V	I_L = 5mA, 1.5A	T_A = 25°C	-1.275	-1.225	V
				-55°C ≤ T_A ≤ 125°C	-1.30	-1.20	V
		V_{IN} =-41.25V	I_L = 5mA, 200mA	T_A = 25°C	-1.275	-1.225	V
				-55°C ≤ T_A ≤ 125°C	-1.30	-1.20	V
		V_{IN} = -6.25V	I_L = 5mA	T_A = 150°C	-1.30	-1.20	V
Line regulation	V_{RLINE}	-41.25V ≤ V_{IN} ≤ -4.25V	I_L = 5mA	T_A = 25°C	-9	9	mV
				-55°C ≤ T_A ≤ 125°C	-23	23	mV
Load regulation	V_{RLOAD}	V_{IN} = -6.25V	5mA ≤ I_L ≤ 1.5A	T_A = 25°C	-6	6	mV
				-55°C ≤ T_A ≤ 125°C	-12	12	mV
		V_{IN} =-41.25V	5mA ≤ I_L ≤ 150mA	T_A = 25°C	-6	6	mV
				-55°C ≤ T_A ≤ 125°C	-12	12	mV
Thermal regulation	V_{RTH}	V_{IN} = -14.6V	I_L = 1.5A	T_A = 25°C	-5	5	mV
Adjust pin current	I_{ADJ}	V_{IN} = -4.25V	I_L = 5mA		25	100	µA
		V_{IN} = -41.25V	I_L = 5mA		25	100	µA
Adjust pin current change versus line voltage	ΔI_{ADJ} (LINE)	-41.25V ≤ V_{IN} ≤ -4.25V	I_L = 5mA		-5	5	µA
Adjust pin current change versus load current	ΔI_{ADJ} (LOAD)	V_{IN} = -6.25V	5mA ≤ I_L ≤ 1.5A		-5	5	µA
Minimum load current	I_Q	-14.25V ≤ V_{IN} ≤ -4.25V forced V_{OUT} = -1.4V			0.20	3.00	mA
		V_{IN} = -41.25V forced V_{OUT} = -1.4V			1.00	5.00	mA
Output short circuit current	I_{OS1}	V_{IN} = -4.25V			1.5	3.5	A
	I_{OS2}	V_{IN} = -40V			0.2	0.8	A
Output voltage recovery after output short circuit current	V_{OUT} (RECOV)	V_{IN} = -4.25V	R_L = 0.833Ω, C_L = 20 µF	after I_{OS1} 3/	-1.30	-1.20	V
		V_{IN} = -40V	R_L = 250Ω	after I_{OS2} 3/	-1.30	-1.20	V
Voltage start-up	V_{START}	V_{IN} = -4.25V	R_L = 0.833Ω, C_L = 20µF		-1.30	-1.20	V
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = -6.25V e_i = 1 Vrms @ fo =2400 Hz	I_L = 500mA	T_A = 25°C 4/ (figure 13)	50	-	dB
Output noise voltage	V_{NO}	V_{IN} = -6.25V	I_L = 100mA	T_A = 25°C 4/ (figure 14)	-	120	µVrms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = -6.25V ΔV_{IN} = -1.0V	I_L = 10mA	T_A = 25°C 5/ (figure 15)	-	40	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = -6.25V	I_L = 100mA ΔI_L = 400mA	T_A = 25°C 5/ (figure 16)	-	0.15	mV/mA

1/ All tests performed at T_A = 125°C may, at the manufacturer's option, be performed at T_A = 150°C. Specifications for T_A = 125°C shall then apply at T_A = 150°C.

2/ Static tests with load currents greater than 5 mA are performed under pulsed conditions defined on figure 12.

3/ Output voltage recovery test shall be performed, with the designated load conditions, immediately after removal of each I_{OS} test forced output voltage condition.

4/ The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

5/ The oscilloscope shall have a bandwidth between 5 and 15 MHz.

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3.3.1 Lead material. The lead material shall be in accordance with MIL-M-38510, or as follows:

Type X

Nickel	50-53 percent
Manganese	0.60 percent, maximum
Silicon	0.30 percent, maximum
Carbon	0.10 percent, maximum
Chromium	0.25 percent, maximum
Cobalt	0.50 percent, maximum
Phosphorous	0.025 percent, maximum
Sulfur	0.025 percent, maximum
Aluminum	0.10 percent, maximum
Iron	Remainder

3.3.2 Lead finish. The lead finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating ambient temperature range of -55°C to +125°C, unless otherwise specified in table I (see table I).

3.4.1 Stability. If the device is located an appreciable distance from the power supply filter, a solid tantalum bypass capacitor should be connected as close to the device V_{CC} input as possible to suppress oscillation. A solid tantalum bypass capacitor is recommended on the device output. Output current oscillations may occur in the current limit operating mode. Since load currents of less than 5 milliamperes may result in a loss of voltage regulation, regulators should be preloaded with 5 milliamperes of load current in lightly loaded applications.

3.4.2 Test limits. The test limits specified in tables I and III apply only for the stated test conditions (e.g., 2 percent duty cycle), which essentially keep the junction temperature constant. In most applications the junction temperature will greatly exceed the 25°C ambient or sink temperature; thus devices may not perform within the 25°C specified limits.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III and limits of table IV which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

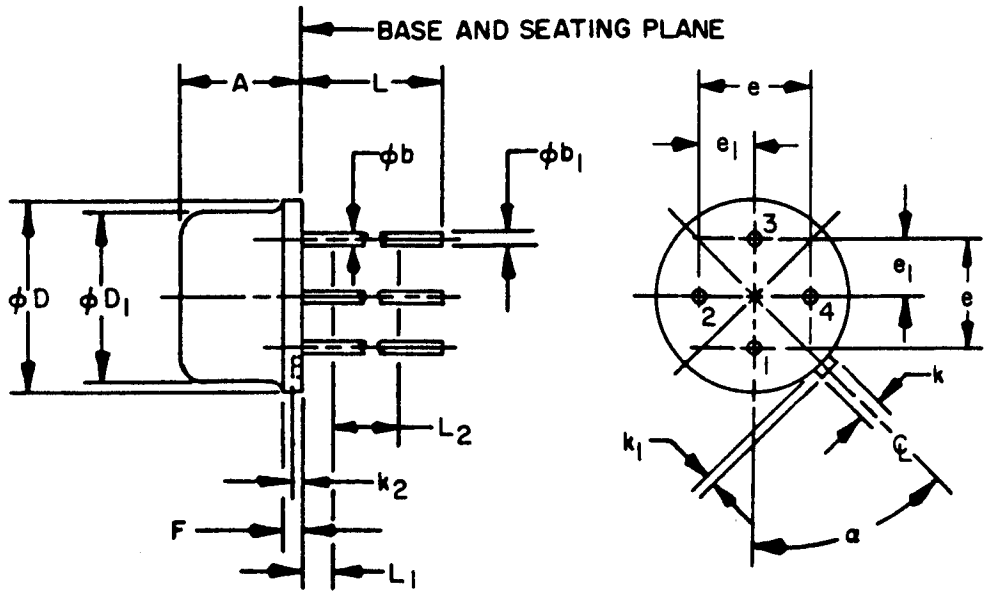
TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,4	1*,2,3,4	1,4
Group A test requirements (method 5005)	1,2,3,4,7	1,2,3,4,7	1,2,3,4,7
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	Not applicable	None	None
Group D end-point electrical parameters (method 5005)	1,2,3	1	1

* PDA applies to subgroup 1 (see 4.2e).

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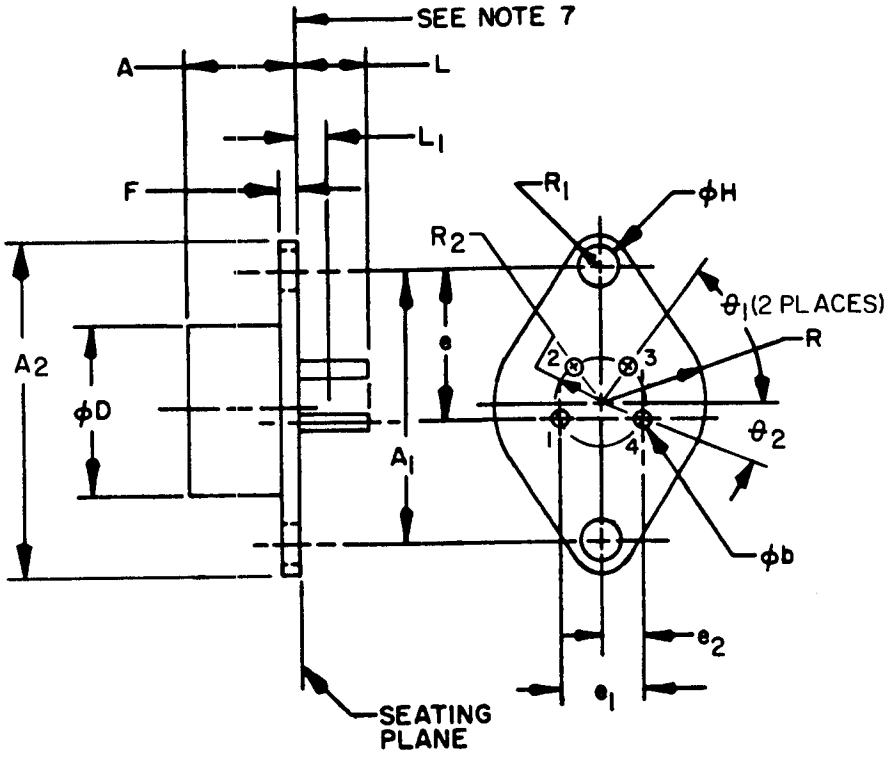
Dimensions

Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.240	.260	6.10	6.60	
Øb	.016	.019	.41	.48	3
$Øb_1$	.016	.021	.41	.53	3
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.79	8.51	
e	.200 T.P.		5.08 T.P.		5
e_1	.100 T.P.		2.54 T.P.		5
F	---	.050	---	1.27	
k	.028	.034	.71	.86	
k_1	.029	.045	.74	1.14	4
k_2	.009	.041	.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Four leads) Øb applies between L_1 and L_2. $Øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Four leads.
5. Measured from the maximum diameter of the product.
6. Leads having a maximum diameter .019 (.48 mm) measured in gaging plane .054 (1.37) + .001 (.03 mm) - .000 (.00 mm) below the base plane of the product shall be within .007 (.18 mm) of their true position relative to a maximum width tab.
7. The product may be measured by direct methods or by gage.

FIGURE 1. Case outline X (device type 01).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
A_2	1.480	1.500	37.59	38.10	
Øb	.038	.043	.97	1.09	3,7
ØD	---	.875	---	22.22	
e	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
ØH	.151	.161	3.84	4.09	5,6
L	.312	.500	7.92	12.70	4
L_1	---	.050	---	1.27	3,4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
R_2	.470 T.P.		11.94 T.P.		
θ_1	54° T.P.		54° T.P.		
θ_2	18° T.P.		18° T.P.		

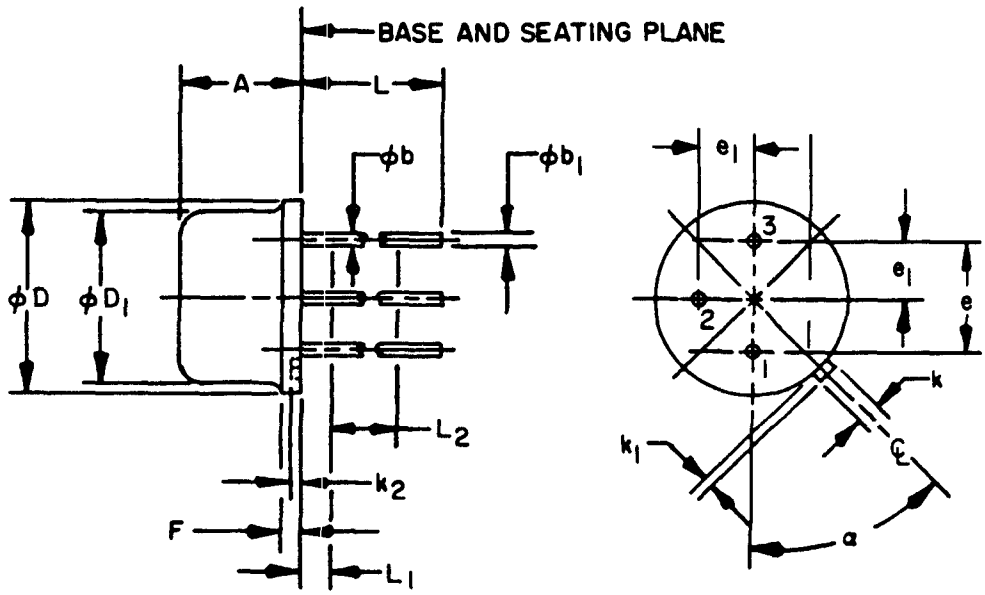
FIGURE 2. Case outline Y (device type 02).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Four leads) Øb applies between L_1 and .500 (12.70 mm) from the seating plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Four leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (.25 mm).
7. Leads having a maximum diameter of .043 (1.09 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) -.000 (.00 mm) below the seating plane shall be located at true position within diameter .014 (.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (.15 mm) overall.

FIGURE 2. Case outline Y (device type 02) - Continued.

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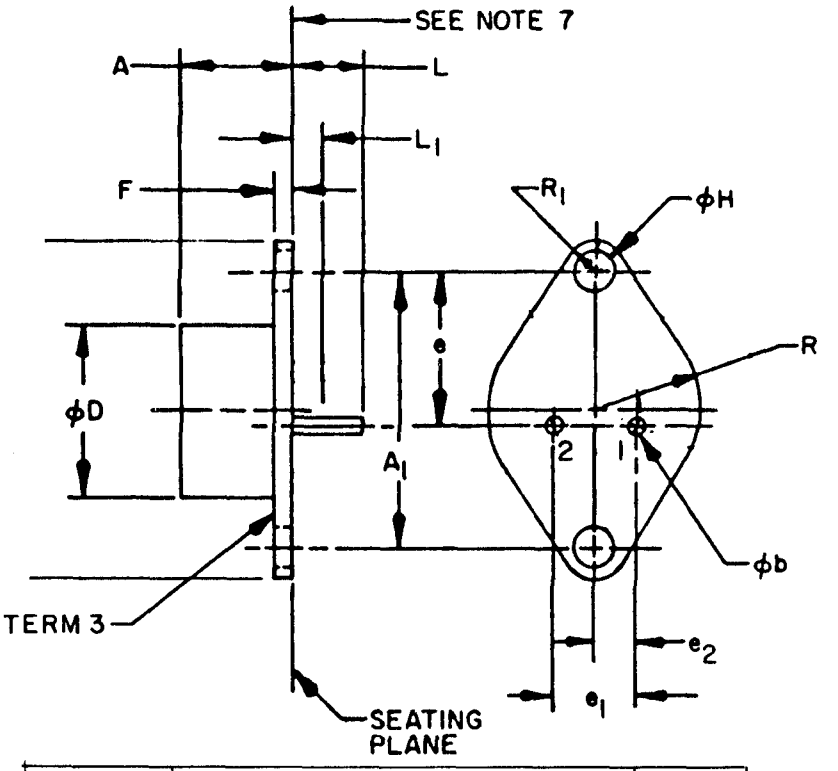


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.165	.185	4.19	4.70	
Øb	.016	.019	.41	.48	3
$Øb_1$	.016	.021	.41	.53	3
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
e	.200 T.P.		5.08 T.P.		5
e_1	.100 T.P.		2.54 T.P.		5
F	---	.050	---	1.27	
k	.028	.034	.71	.86	
k_1	.029	.045	.74	1.14	4
k_2	.009	.041	.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Three leads) Øb applies between L_1 and L_2. $Øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Three leads.
5. Measured from the maximum diameter of the product.
6. Leads having a maximum diameter .019 (.48 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) - .000 (.00 mm) below the base plane of the product shall be within .007 (.18 mm) of their true position relative to a maximum width tab.
7. The product may be measured by direct methods or by gage.

FIGURE 3. Case outline X (device type 03).

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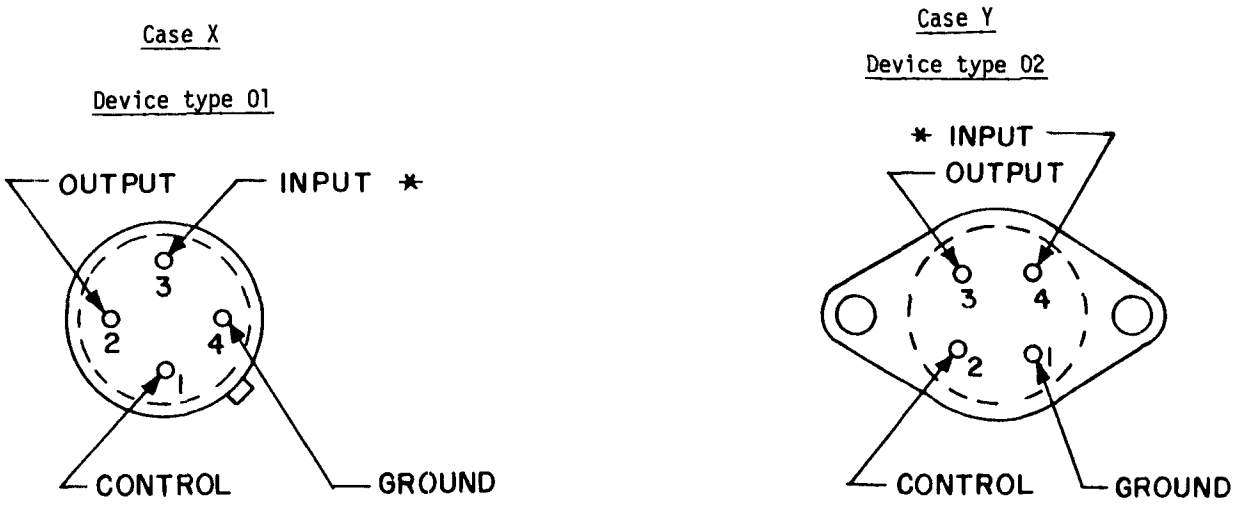
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
Øb	.038	.043	.97	1.09	3,7
ØD	---	.875	---	22.22	
e	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
ØH	.151	.161	3.84	4.69	5,6
L	.312	.500	7.92	12.70	4
L_1	---	.050	---	1.27	3,4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. (Two leads) Øb applies between L_1 and .500 (12.70 mm) from the seating plane.

   Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Two leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (.25 mm).
7. Leads having a maximum diameter of .043 (1.09 mm) measured in gaging plane .054 (1.37 mm) + .001 (.03 mm) - .000 (.00 mm) below the seating plane shall be located at true position within diameter .014 (.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (.15 mm) overall.

FIGURE 4. Case outline Y (device type 04).

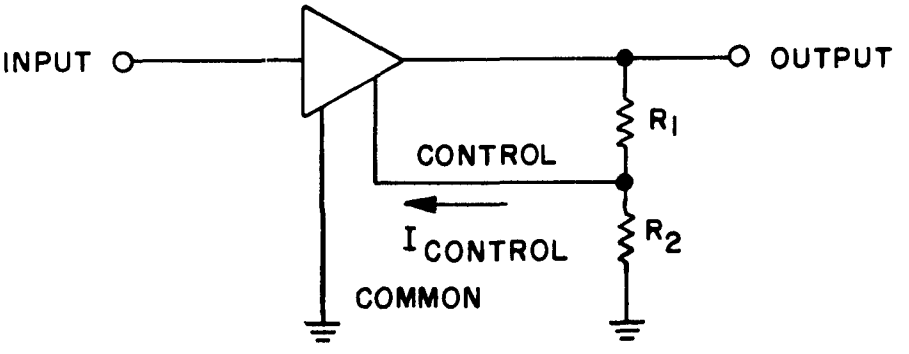
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NOTE: Bottom views.

*(Case is connected to input)

FIGURE 5. Terminal connections for device types 01 and 02.



NOTES:

1. $V_{OUT} = \left[\left(\frac{R_1 + R_2}{R_2}\right)\left(V_{CONTROL}\right) + \left|I_{CONTROL}\right| \times R_2\right]$ Volts
2. $V_{CONTROL}$ = -2.23 V (nominal).
3. R_2 = 446Ω provides a minimum of |5 mA| load to the regulator at any V_{OUT}.

FIGURE 6. Block diagram for device types 01 and 02.

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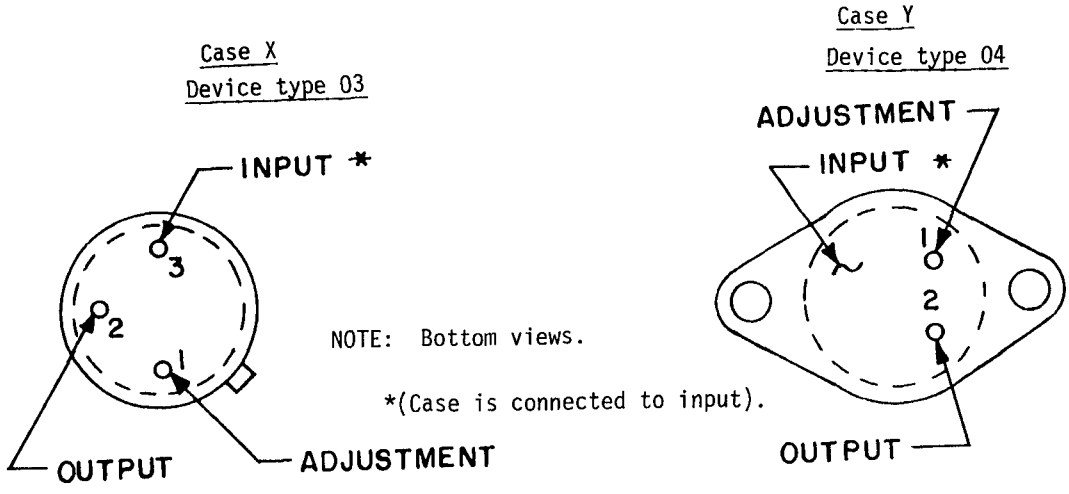
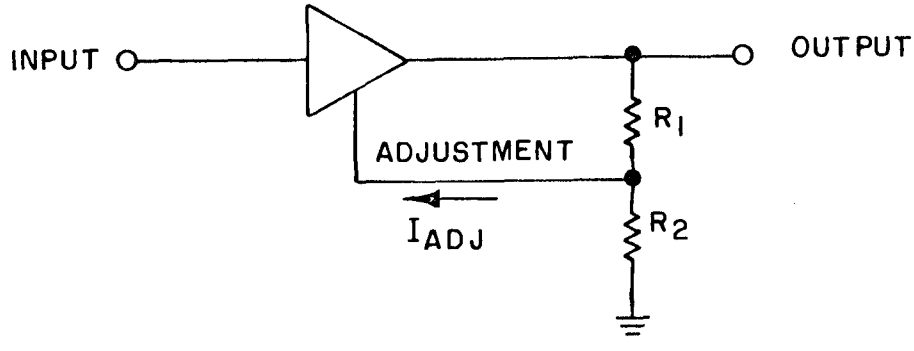


FIGURE 7. Terminal connections for 03 and 04.



NOTES:

1. $V_{OUT} = \left[\left(\frac{R_1 + R_2}{R_1}\right)\left(-1.25\right) + \left|I_{ADJ}\right| x\ R_2\right]$ Volts.
2. $R_1 = 250\Omega$.

FIGURE 8. Block diagram for device types 03 and 04.

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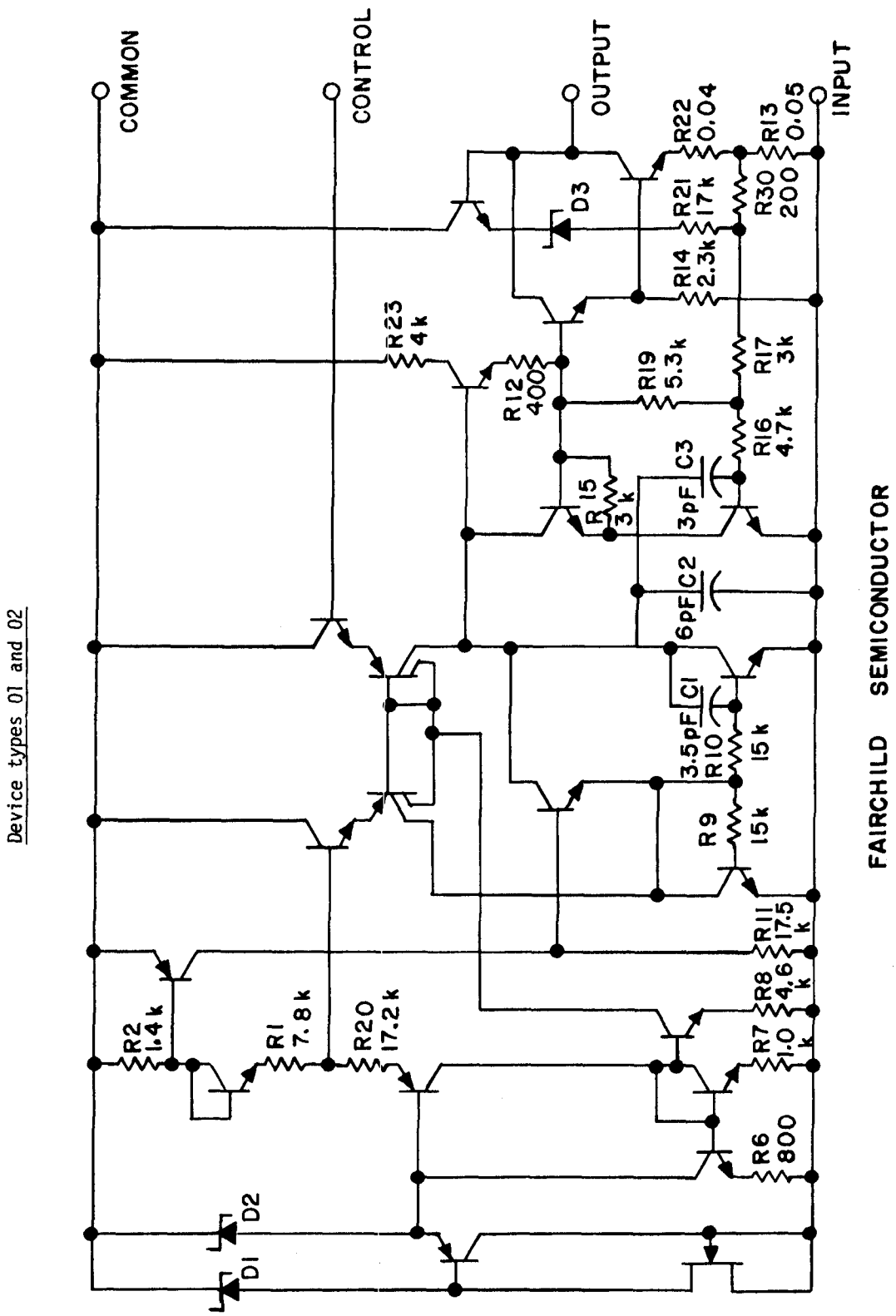


FIGURE 9. Schematic circuits.

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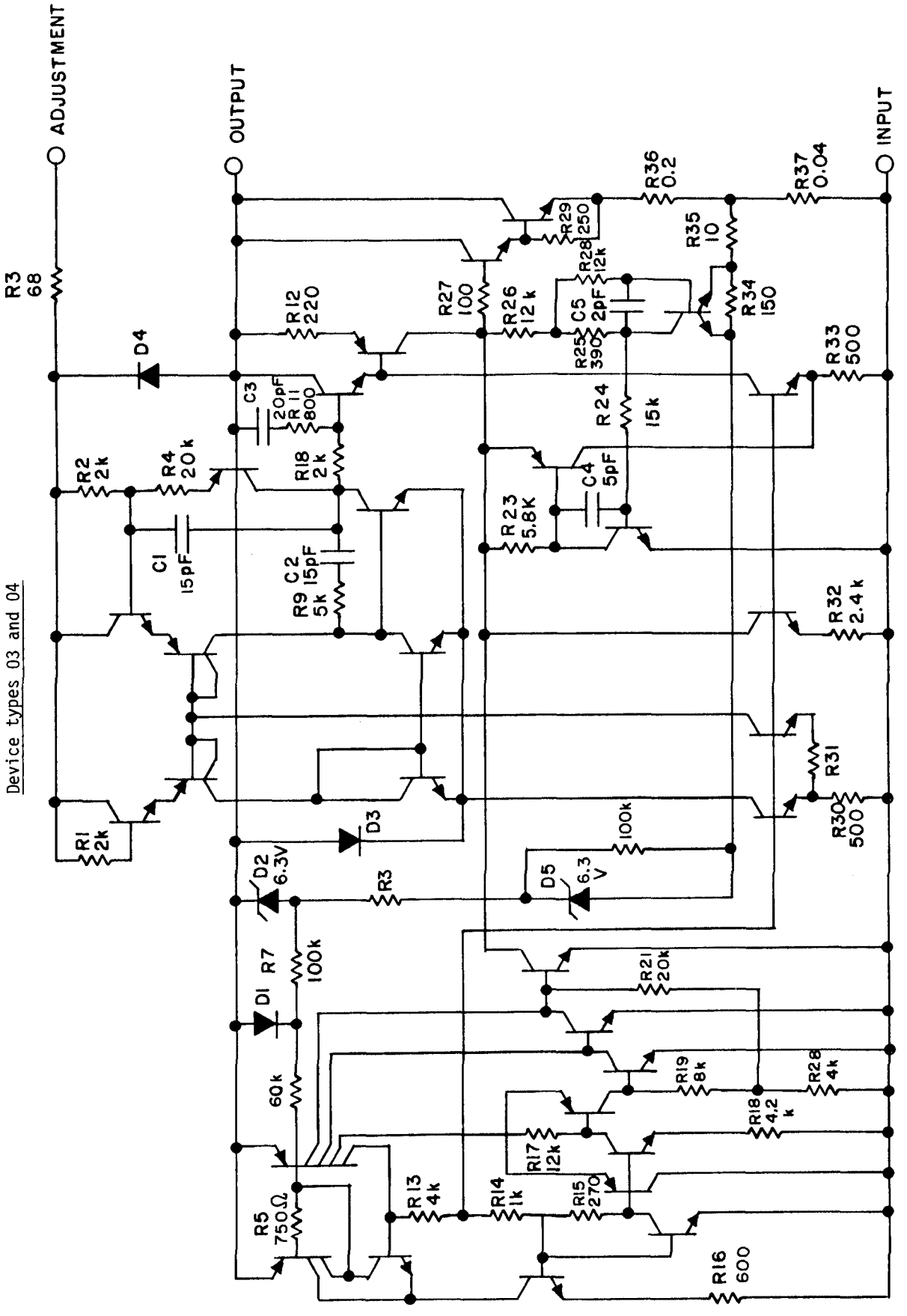
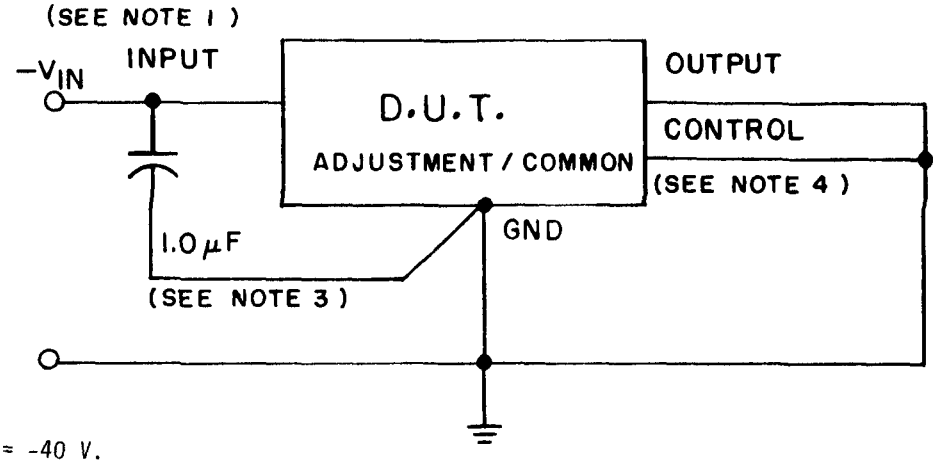


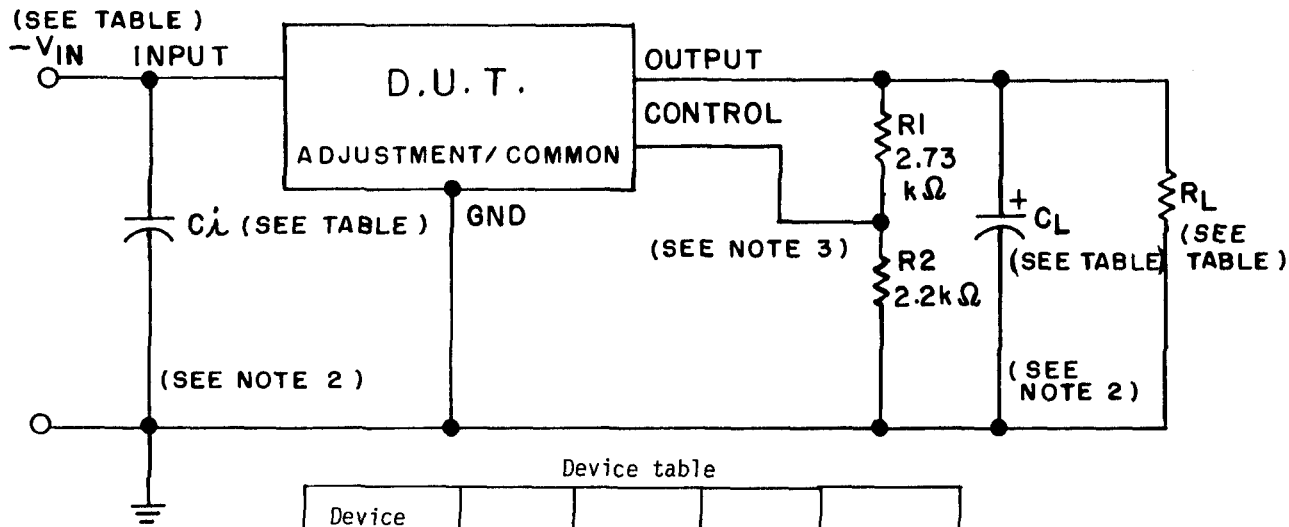
FIGURE 9. Schematic circuits - Continued.

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NOTES:
1. V_{IN} = -40 V.
2. Test is conducted at T_A = 25°C without a heat sink for a minimum of 4 hours.
3. The 1.0 μF capacitor may be reduced or eliminated provided the device remains stable.
4. The control pin connection is required for device types 01 and 02 only.

FIGURE 10. Pre burn-in test circuit.



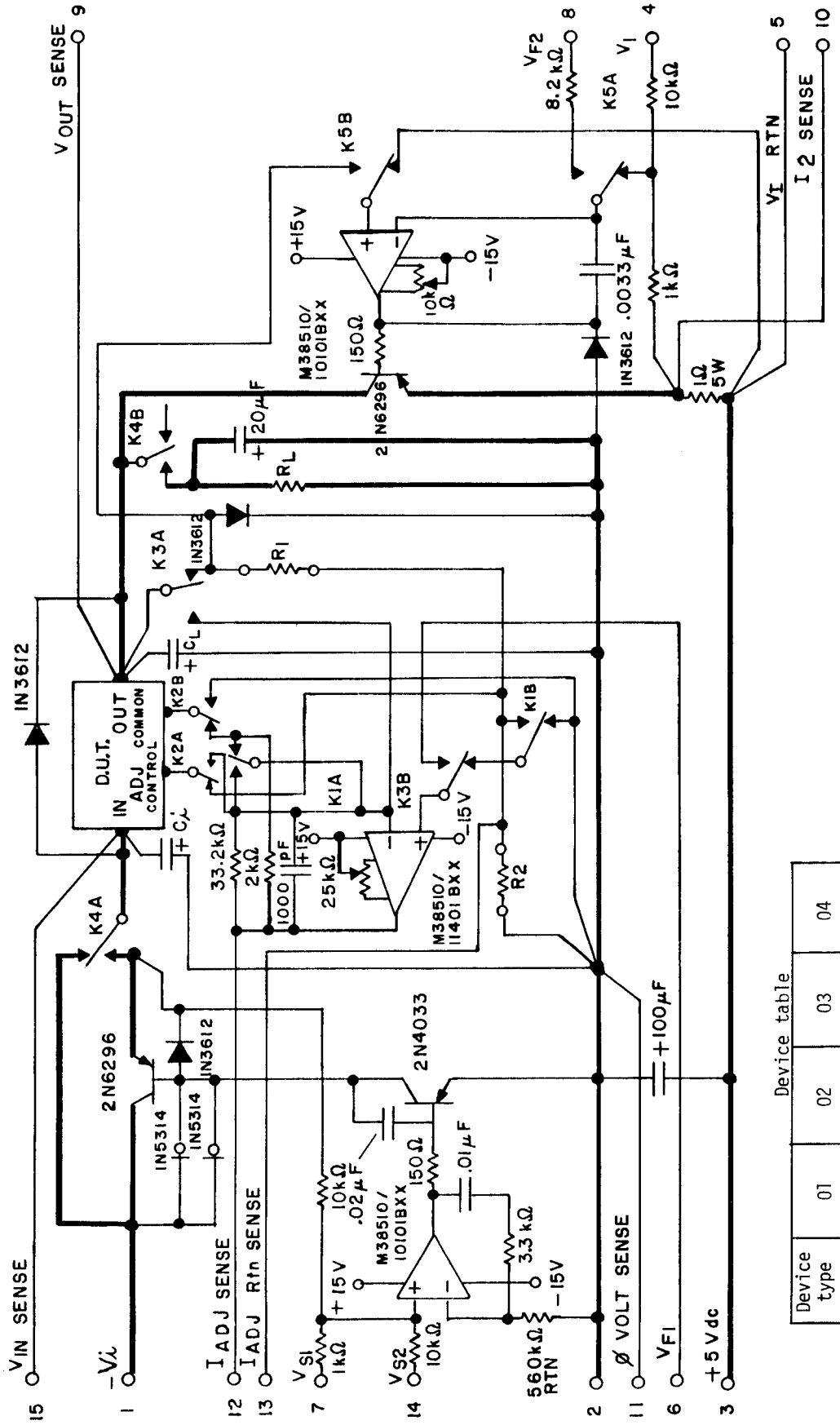
Device table

Device type	01	02	03	04
$-V_{IN}$	-20 V	-20 V	-36.5 V	-36.5 V
R_L	887Ω	118Ω	250Ω	63.4Ω
C_i and C_L	2.0 μF	2.0 μF	1.0 μF	1.0 μF

NOTES:
1. Test is conducted without a heat sink.
2. The capacitors may be reduced or eliminated provided the device remains stable.
3. The control pin connection and resistors (R_1 and R_2) are required for device types 01 and 02 only.

FIGURE 11. Burn-in and operating life test circuit.

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Device table

Device type	01	02	03	04
R_1	549Ω	549Ω	249Ω	249Ω
R_2	442Ω	442Ω	0	0
R_L	10Ω	5Ω	2.5Ω	0.833Ω
C_i, C_L	2.0 μF	2.0 μF	1.0 μF	1.0 μF

FIGURE 12. Test circuit for static tests.

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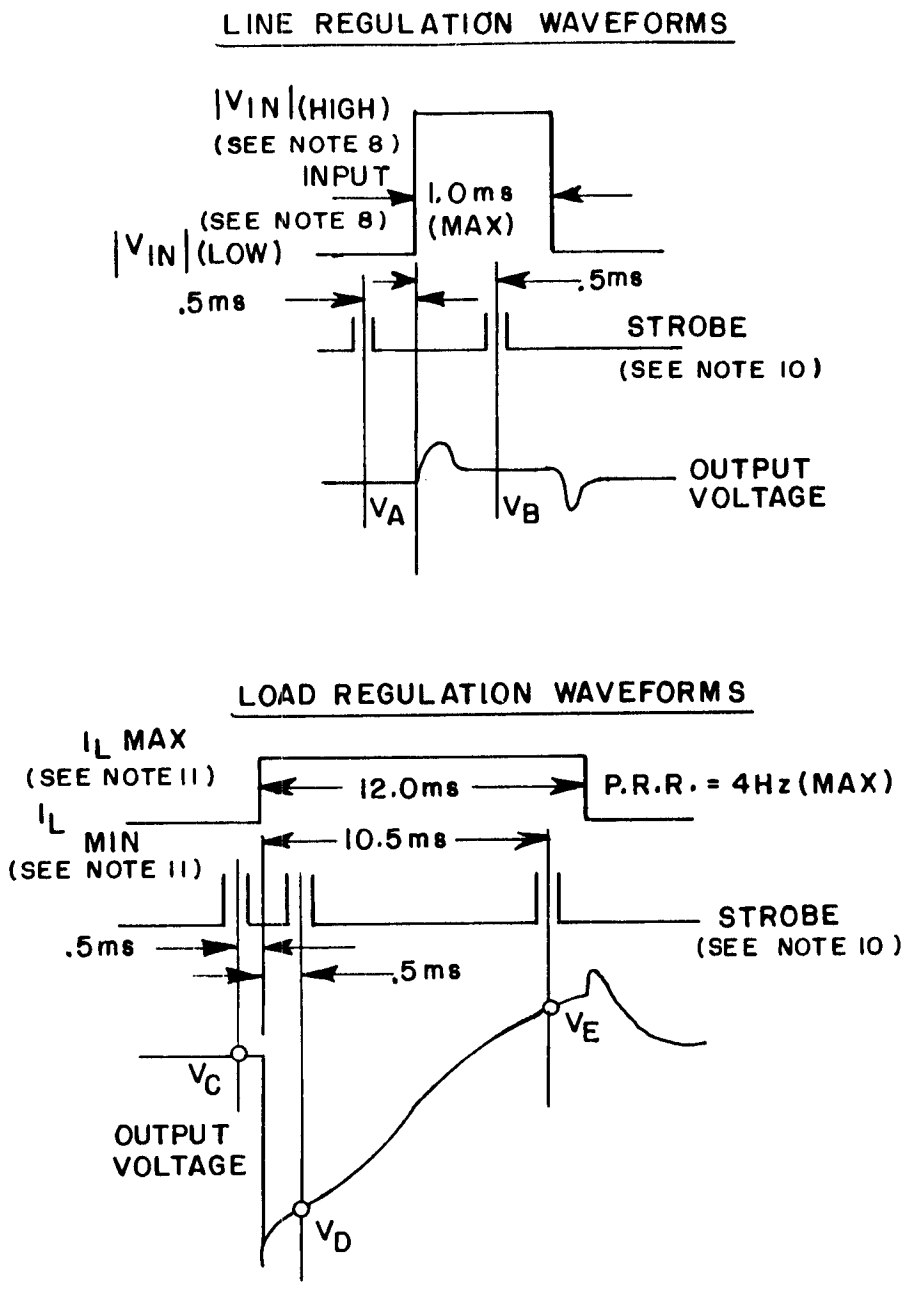


FIGURE 12. Test circuit for static tests - Continued.

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WAVEFORMS
FOR
VOLTAGE START-UP,
OUTPUT SHORT CIRCUIT CURRENT,
AND
OUTPUT VOLTAGE RECOVERY

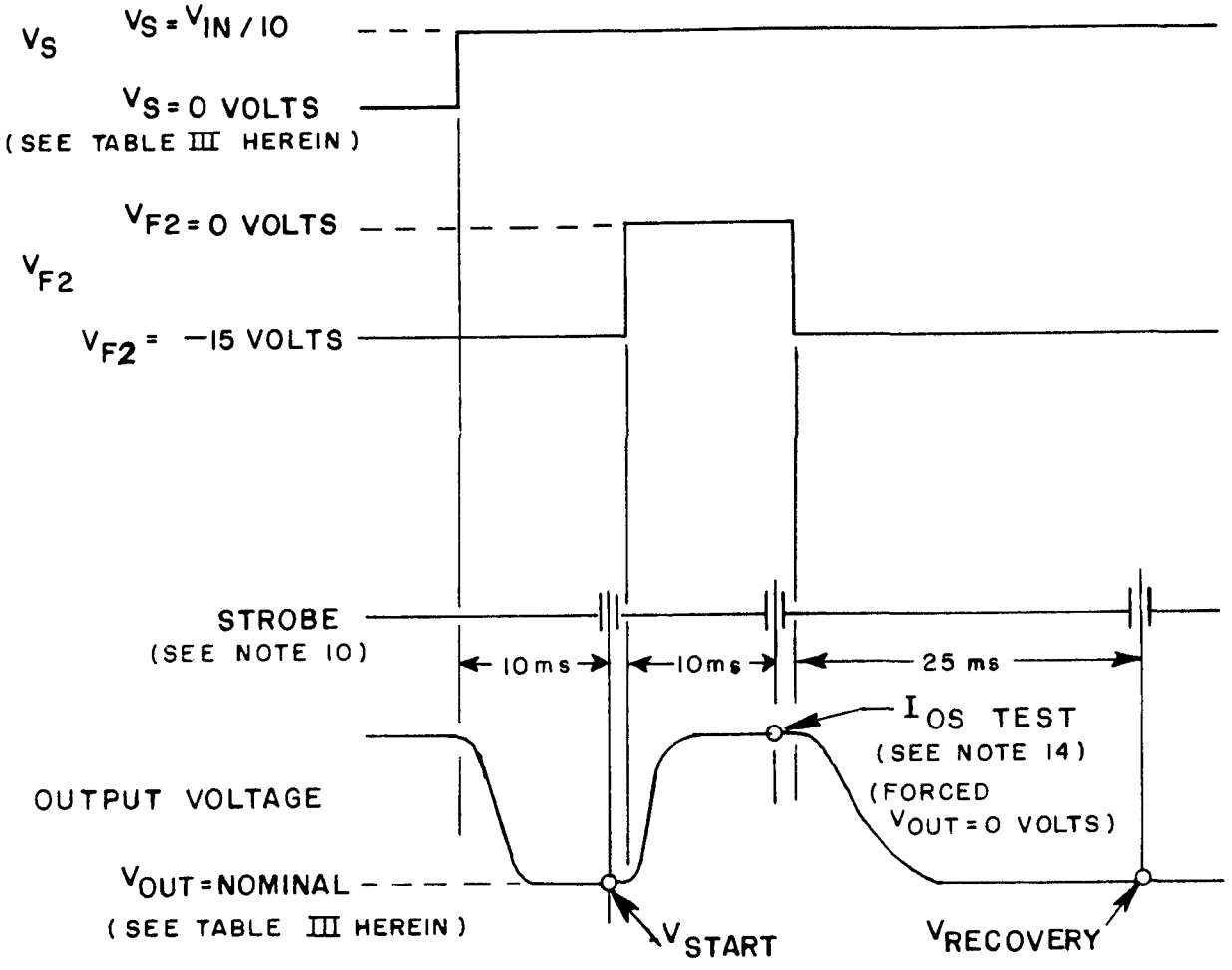


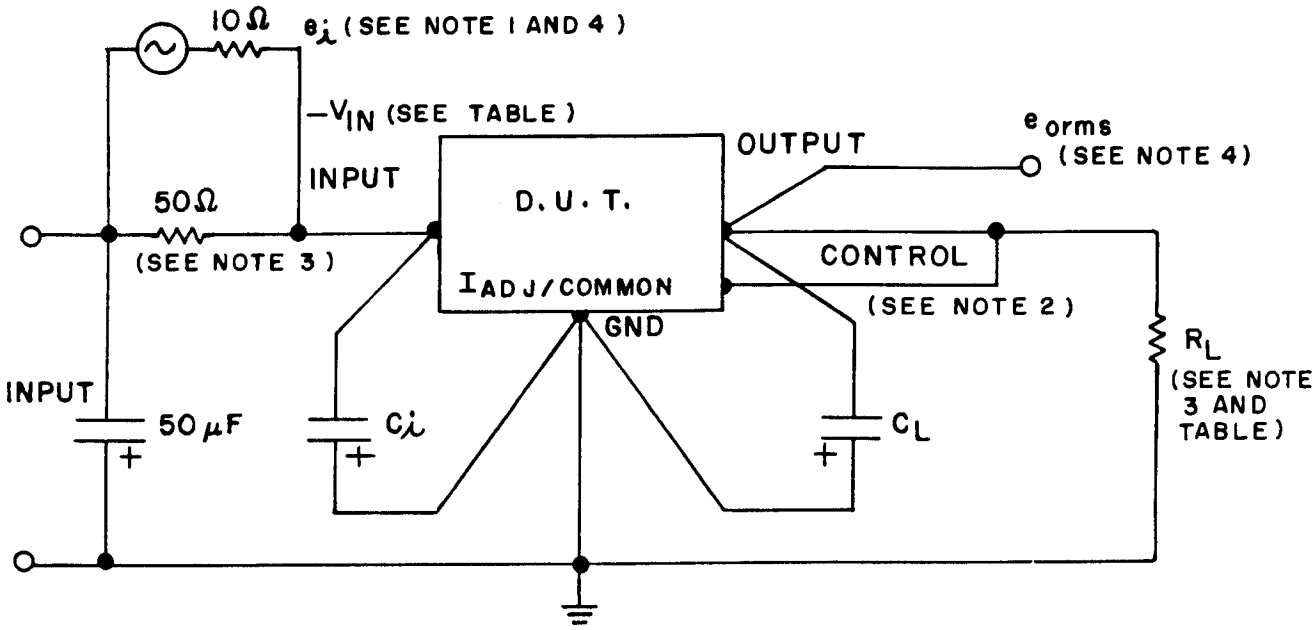
FIGURE 12. Test circuit for static tests - Continued.

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NOTES:
1. Heavy current paths ($I \geq 1.0$ A) are indicated by bold lines.
2. Kelvin connections must be used for all output current and voltage measurements. For device types 03 and 04, output voltage measurements shall be made 1/8 inch below the case.
3. The output offset voltage shall be adjusted to zero with the DUT removed. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6296 may be used to develop the proper load current and input voltage pulses. These circuits shall require the approval of the qualifying activity.
4. Relay switch positions are defined in table III.
5. Load currents of 5 mA may be established via the load resistors R_1 and R_2. All other load currents shall be established via the pulse load circuit. Resistors R_1 and R_2 shall have a tolerance $\leq 0.1\%$ for device types 01 and 02.
6. The pulse generator for the pulse load circuit shall have the following characteristics:
   a. Pulse amplitude = $-10\ (|I_L| - V_o/(R_1 + R_2))$ volts.
   b. Pulse width = 1.0 ms (unless otherwise stated).
   c. Duty cycle = 2% (maximum).
7. Load currents shall be determined by the voltage measured across the 1Ω resistor. Measurements shall be made 0.5 ms after the start of the pulse.
8. $|V_{IN}|$ (LOW) and $|V_{IN}|$ (HIGH) per table III herein.
9. $V_{RLINE} = V_B - V_A$.
10. The output voltage is sampled at specified intervals. Strobe pulse width is 100 µs maximum.
11. I_L (minimum) and I_L (maximum) per table III herein.
12. $V_{RLOAD} = V_D - V_C$.
13. $V_{RTH} = V_D - V_E$.
14. $I_{OS} = (I_L)$ amps.

FIGURE 12. Test circuit for static tests - Continued.

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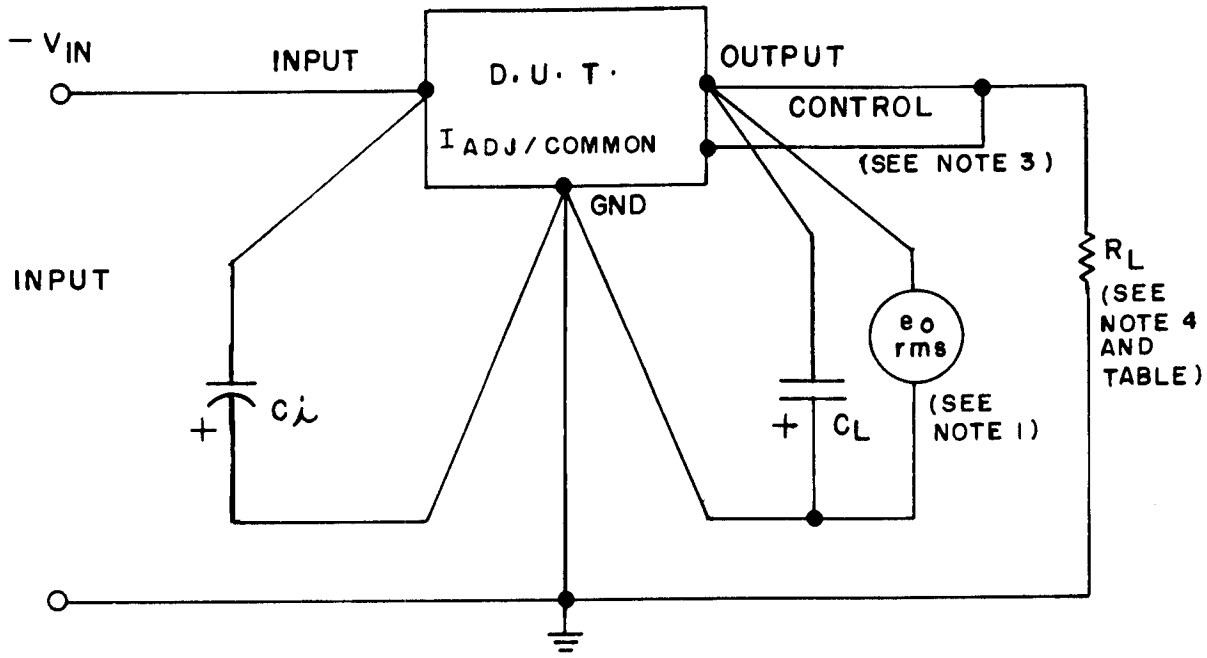
Device table

Device type	01	02	03	04
V_{IN}	-10 V	-10 V	-6.25	-6.25
R_L	40.2Ω	14.3Ω	10Ω	2.5
C_i, C_L	2.0 μF	2.0 μF	1.0 μF	1.0 μF

NOTES:
1. e_i = 1 Vrms @ f = 2400 Hz (measured at the input terminals of the DUT).
   Ripple rejection = 20 log $\frac{e_{irms}}{e_{orms}}$.
2. The control pin connection is required for device types 01 and 02 only.
3. The input 50Ω resistor and R_L shall be type RER 70 or equivalent.
4. The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

FIGURE 13. Ripple rejection test circuit.

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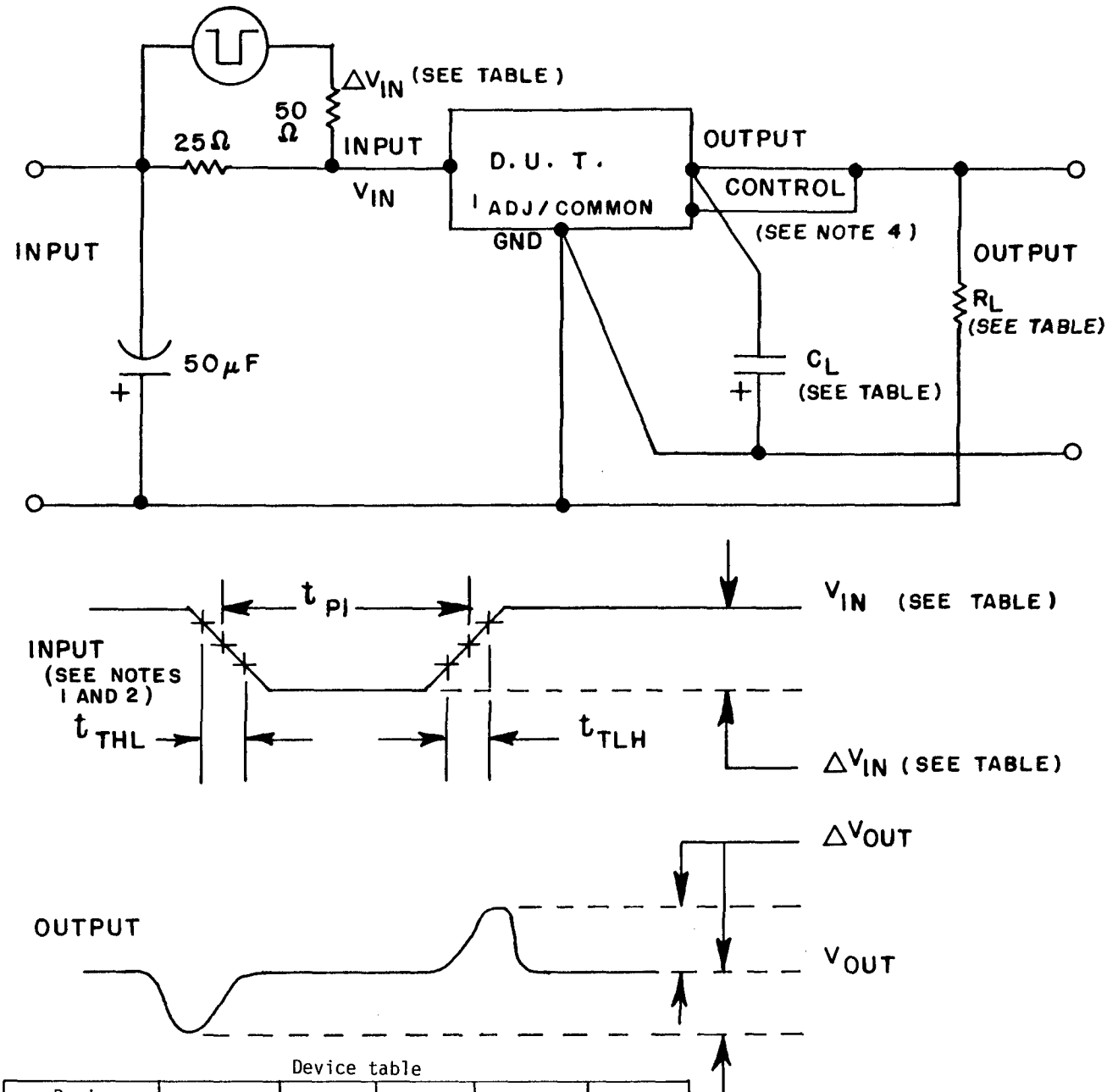
Device table

Device type	01	02	03	04
V_{IN}	10 V	10 V	6.25 V	6.25
R_L	100Ω	50Ω	25Ω	12.5Ω
C_i, C_L	2.0 µF	2.0 µF	1.0 µF	1.0 µF

NOTES:
1. The meter for measuring e_{orms} shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.
2. $N_o = e_o$ rms.
3. The control pin connection is required for device types 01 and 02 only.
4. R_L shall be type RER 70 or equivalent.

FIGURE 14. Noise test circuit.

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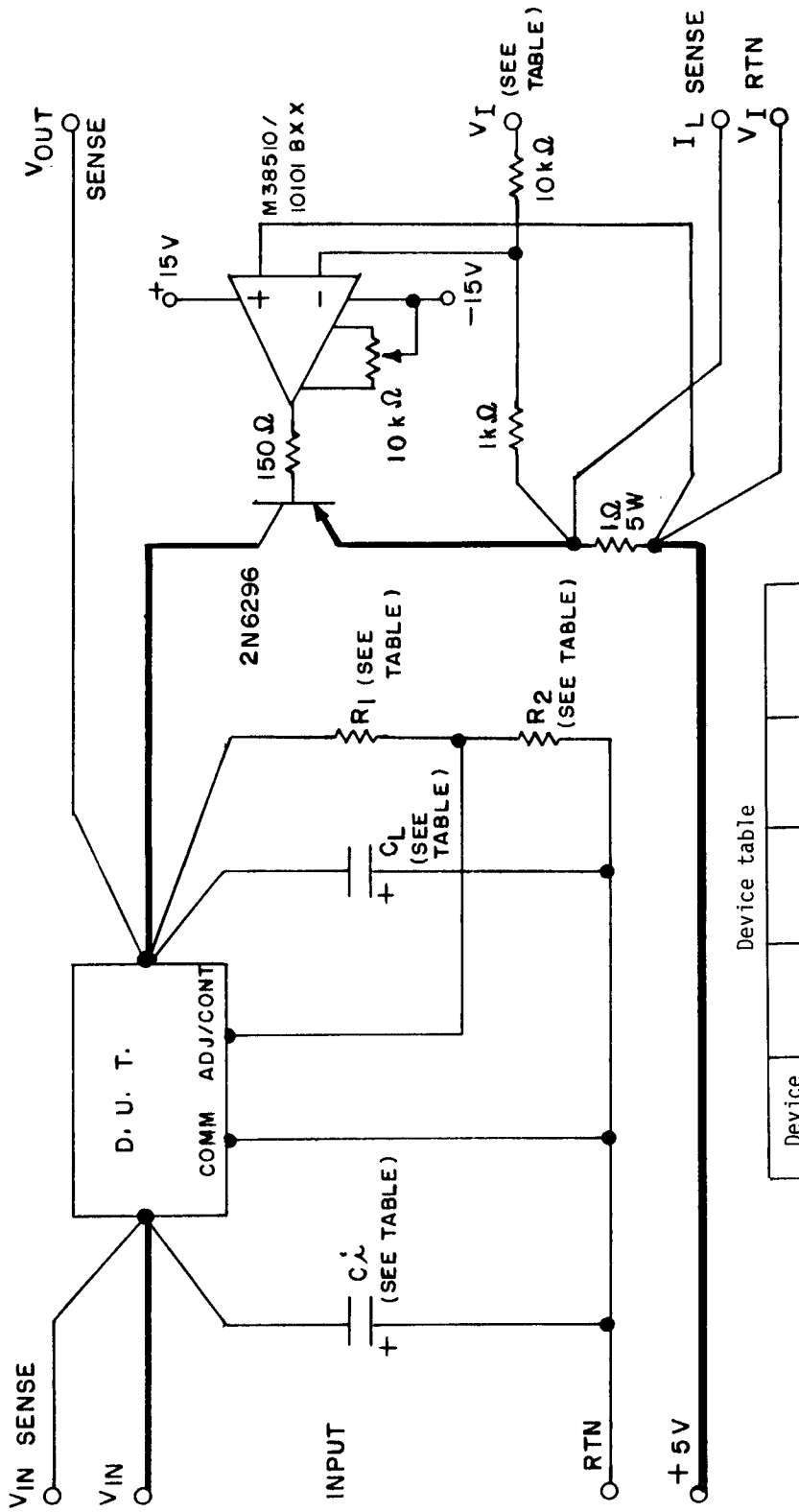
Device table

Device type	01	02	03	04	Notes
V_{IN}	-10 V	-10 V	-6.25 V	-6.25 V	1
ΔV_{IN}	-3.0 V	-3.0 V	-3.0 V	-3.0 V	1
R_L	1.25 kΩ	1.25 kΩ	120Ω	120Ω	
$t_{THL} = t_{TLH}$	5.0 μs	5.0 μs	5.0 μs	5.0 μs	1
C_i, C_L	2.0 μF	2.0 μF	1.0 μF	1.0 μF	

NOTES:
1. Measured at device input.
2. Pulse width t_{pl} = 25 μs; duty cycle = 3% (maximum).
3. Oscilloscope bandwidth = 5 MHz to 15 MHz.
4. The control pin connection is required for device types 01 and 02 only.

FIGURE 15. Line transient response test circuit.

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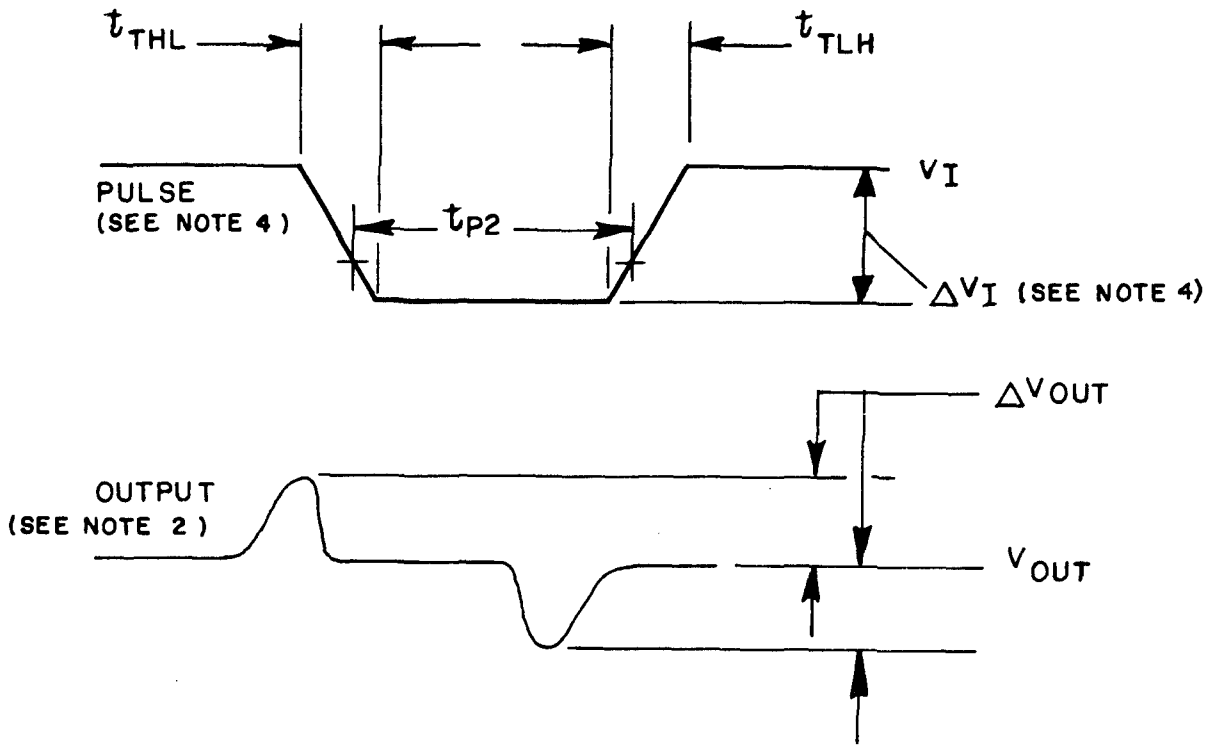


Device table

Device type	01	02	03	04
R_1	0	0	249Ω	249Ω
R_2	5.0 kΩ	5.0 kΩ	0	0
I_L	50 mA	100 mA	50 mA	100 mA
ΔI_L	200 mA	400 mA	200 mA	400 mA
V_I	0.49 V	0.99 V	0.45 V	0.95 V
ΔV_I	2.0 V	4.0 V	2.0 V	4.0 V
C_i, C_L	2.0 µF	2.0 µF	1.0 µF	1.0 µF

FIGURE 16. Load transient response test circuit.

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NOTES:
1. Heavy current paths ($I \geq 1.0$ A) are indicated by bold lines.
2. Kelvin connections must be used for all output current and voltage measurements.
3. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6296 may be used to develop the proper load current and input voltage pulses. These circuits shall require the approval of the qualifying activity.
4. The pulse generator for the pulse load circuit shall have the following characteristics. (See device table.)
   a. Voltage level (V_I) = 10 ($|I_L| - \frac{V_{OUT}}{R_1+R_2}$) volts.
   b. Pulse width (t_{P2}) = 25 µsec.
   c. Duty cycle = 3% (maximum).
   d. $t_{THL} = t_{TLH}$ = 1.0 µsec for device types 01 and 02.
   e. $t_{THL} = t_{TLH}$ = 5.0 µsec for device types 03 and 04.
   f. Difference voltage level (ΔV_I) = 10 (I_L) volts.
5. a. ΔV_{OUT} = 500 mV maximum for device type 01.
   b. ΔV_{OUT} = 1,000 mV maximum for device type 02.
   c. ΔV_{OUT} = 60 mV maximum for device types 03 and 04.
   (These values guarantee the specified limits for load transient response.)
6. The oscilloscope shall have a bandwidth between 5 and 15 MHz.

FIGURE 16. Load transient response test circuit - Continued.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
1 (T_A = 25°C)	V_{OUT1}	1	-8	5	-8	-	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		-5.25	-4.75	V
	V_{OUT2}	2	-8	500	-8	4.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	-30	5	-30	-	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	-30	50	-30	0.45	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE1}	5	-8	50	-8	0.45	-	-	-			E_5		$V_{RLINE1} = E_5 - E_4$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	6	-8	350	-8	3.45	-	-	-			E_6				-	-	
	V_{RLINE2}	7	-25	350	-25	3.45	-	-	-			E_7		$V_{RLINE2} = E_6 - E_7$		-50	50	
	V_{RLOAD1}	8	-10	5	-10	-	-	-	-			E_8			See figure 12 waveforms	-	-	
	V_{RLOAD1}	9	-10	500	-10	4.95	-	-	-			E_9		$V_{RLOAD1} = E_8 - E_9$		-100	100	
	V_{RLOAD2}	10	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_3 - E_4$		-150	150	
	V_{RTH}	11	-15	500	-15	4.95	-	-	-		9-11	E_{10}		$V_{RTH} = E_{10}$	See figure 12 waveforms	-50	50	
	I_{SCD1}	12	-10	5	-10	-	-	-	-		12-13	E_{11}		$I_{SCD1} = \frac{E_{11}}{2000}$		0.1	3.0	mA
	I_{SCD2}	13	-30	5	-30	-	-	-	-			E_{12}		$I_{SCD2} = \frac{E_{12}}{2000}$		0.1	4.0	
	ΔI_{SCD} (LINE)	14	-8	5	-8	-	-	-	-			E_{13}		ΔI_{SCD} (LINE) $= \frac{E_{13} - E_{12}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	15	-10	500	-10	4.95	-	-	-			E_{14}		ΔI_{SCD} (LOAD) $= \frac{E_{11} - E_{14}}{2000}$		-0.5	0.5	
	I_{OS1}	16	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{15}		$I_{OS1} = E_{15}$	See figure 12 waveforms	0.002	2.0	A
	V_{OUT5} (RECOV)	17	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{16}		$V_{OUT5} = E_{16}$		-5.25	-4.75	V
	I_{OS2}	18	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{17}		$I_{OS2} = E_{17}$		0.002	1.0	A
	V_{OUT6} (RECOV)	19	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{18}		$V_{OUT6} = E_{18}$		-5.25	-4.75	V
	I_{CTL}	20	-10	350	-10	3.45	-	-	-	K1,K2	12-13	E_{19}		$I_{CTL} = \frac{E_{19}}{33200}$		0.001	2	µA
	V_{START}	21	-8	500	-15	-	-	-0.8	-	K4	9-11	E_{20}		$V_{OUT} = E_{20}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	22	-38	500	-38	4.95	-	-	-	None		E_{21}		$V_{OUT7} = E_{21}$	R_1 = 5.49kΩ ±0.1%	-31.5	-28.5	
2 (T_A = 125°C)	V_{OUT1}	23	-8	5	-8	-	-	-	-			E_{22}		$V_{OUT1} = E_{22}$		-5.25	-4.75	
	V_{OUT2}	24	-8	500	-8	4.95	-	-	-			E_{23}		$V_{OUT2} = E_{23}$				
	V_{OUT3}	25	-30	5	-30	-	-	-	-			E_{24}		$V_{OUT3} = E_{24}$				
	V_{OUT4}	26	-30	50	-30	0.45	-	-	-			E_{25}		$V_{OUT4} = E_{25}$				
	V_{RLINE1}	27	-8	50	-8	0.45	-	-	-			E_{26}		$V_{RLINE1} = E_{26} - E_{25}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	28	-8	350	-8	3.45	-	-	-			E_{27}				-	-	
	V_{RLINE2}	29	-25	350	-25	3.45	-	-	-			E_{28}		$V_{RLINE2} = E_{27} - E_{28}$		-50	50	
	V_{RLOAD1}	30	-10	5	-10	-	-	-	-			E_{29}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	31	-10	500	-10	4.95	-	-	-			E_{30}		$V_{RLOAD1} = E_{29} - E_{30}$		-100	100	
	V_{RLOAD2}	32	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{24} - E_{25}$		-150	150	
	I_{SCD1}	33	-10	5	-10	-					12-13	E_{31}		$I_{SCD1} = \frac{E_{31}}{2000}$		0.1	3.0	mA
	I_{SCD2}	34	-30	5	-30	-	-	-	-		12-13	E_{32}		$I_{SCD2} = \frac{E_{32}}{2000}$		0.1	4.0	mA

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	See figure 12 — Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) [Hi-Lo pin potential] 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines — Pins	Value	Unit	Equation	Notes	Limits — Min	Max	Unit
2 (T_A = 125°C)	ΔI_{SCD} (LINE)	35	-8	5	-8	-	-	-	-	None	12-13	E_{33}	V	ΔI_{SCD} (LINE) = $\frac{E_{33}-E_{32}}{2000}$		-1.0	1.0	mA
	ΔI_{SCD} (LOAD)	36	-10	500	-10	4.95	-	-	-	None	12-13	E_{34}		ΔI_{SCD} (LOAD) = $\frac{E_{34}-E_{34}}{2000}$		-0.5	0.5	mA
	I_{OS1}	37	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{35}		$I_{OS1} = E_{35}$	See figure 12 waveforms	0.002	2.0	A
	V_{OUT5} (RECOV)	38	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{36}		$V_{OUT5} = E_{36}$		-5.25	-4.75	V
	I_{OS2}	39	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{37}		$I_{OS2} = E_{37}$		0.002	1.0	A
	V_{OUT6} (RECOV)	40	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{38}		$V_{OUT6} = E_{38}$		-5.25	-4.75	V
	I_{CTL}	41	-10	350	-10	3.45	-	-	-	K1,K2	12-13	E_{39}		$I_{CTL} = \frac{E_{39}}{33200}$		0.001	3	µA
	V_{START}	42	-8	500	-15	-	-	-0.8	-	K4	9-11	E_{40}		$V_{OUT} = E_{40}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	43	-38	500	-38	4.95	-	-	-	None		E_{41}		$V_{OUT7} = E_{41}$	R_1 = 5.49 kΩ ±0.1%	-31.5	-28.5	
(T_A = 150°C)	V_{OUT8}	44	-10	5	-10	-	-	-	-			E_{42}		$V_{OUT8} = E_{42}$		-5.30	-4.70	
3 (T_A = -55°C)	V_{OUT1}	45	-8	5	-8	-	-	-	-			E_{43}		$V_{OUT1} = E_{43}$		-5.25	-4.75	
	V_{OUT2}	46	-8	500	-8	4.95	-	-	-			E_{44}		$V_{OUT2} = E_{44}$				
	V_{OUT3}	47	-30	5	-30	-	-	-	-			E_{45}		$V_{OUT3} = E_{45}$				
	V_{OUT4}	48	-30	50	-30	0.45	-	-	-			E_{46}		$V_{OUT4} = E_{46}$				
	V_{RLINE1}	49	-8	50	-8	0.45	-	-	-			E_{47}		$V_{RLINE1} = E_{47}-E_{46}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	50	-8	350	-8	3.45	-	-	-			E_{48}				-	-	
	V_{RLINE2}	51	-25	350	-25	3.45	-	-	-			E_{49}		$V_{RLINE2} = E_{48}-E_{49}$		-50	50	
	V_{RLOAD1}	52	-10	5	-10	-	-	-	-			E_{50}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	53	-10	500	-10	4.95	-	-	-			E_{51}		$V_{RLOAD1} = E_{50}-E_{51}$		-100	100	
	V_{RLOAD2}	54	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{45}-E_{46}$		-150	150	
	I_{SCD1}	55	-10	5	-10	-	-	-	-		12-13	E_{52}		$I_{SCD1} = \frac{E_{52}}{2000}$		0.1	3.0	mA
	I_{SCD2}	56	-30	5	-30	-	-	-	-			E_{53}		$I_{SCD2} = \frac{E_{53}}{2000}$		0.1	4.0	
	ΔI_{SCD} (LINE)	57	-8	5	-8	-	-	-	-			E_{54}		ΔI_{SCD} (LINE) = $\frac{E_{54}-E_{53}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	58	-10	500	-10	4.95	-	-	-			E_{55}		ΔI_{SCD} (LOAD) = $\frac{E_{52}-E_{55}}{2000}$		-0.5	0.5	
	I_{OS1}	59	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{56}		$I_{OS1} = E_{56}$	See figure 12 waveforms	0.002	2.0	A
	V_{OUT5} (RECOV)	60	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{57}		$V_{OUT5} = E_{57}$		-5.25	-4.75	V
	I_{OS2}	61	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{58}		$I_{OS2} = E_{58}$		0.002	1.0	A
	V_{OUT6} (RECOV)	62	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{59}		$V_{OUT6} = E_{59}$		-5.25	-4.75	V
	I_{CTL}	63	-10	350	-10	3.45	-	-	-	K1,K2	12-13	E_{60}		$I_{CTL} = \frac{E_{60}}{33200}$		0.001	3	µA

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
3 (T_A = -55°C)	V_{START}	64	-8	500	-15	-	-	-0.8	-	K4	9-11	E_{61}	V	$V_{OUT} = E_{61}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	65	-38	500	-38	4.95	-	-	-	None	9-11	E_{62}	V	$V_{OUT7} = E_{62}$	R_1 = 5.49 kΩ ± 0.1%	-31.5	-28.5	V

Subgroup	Symbol	Test no.	Test conditions		Measurement sense lines			Equation	Notes	Limits		Unit
			Input voltage	Load current	Symbol	Value	Unit			Min	Max	
4 (T_A = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = -10V e_i = 1.0Vrms @ 2400 Hz	I_L = 125mA	e_{orms}	E_{63}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20 \log E_{63}$	See figure 13	45	-	dB
7 (T_A = 25°C)	V_{NO}	67	V_{IN} = -10V	I_L = 50mA	e_{orms}	E_{64}	Vrms	$V_{NO} = E_{64}$	See figure 14	-	250	μVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = -10V V_{pulse} = -3.0V	I_L = 5mA	V_{OUT}	E_{65}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{65}}{3}$	See figure 15	-	30	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = -10V	I_L = 50mA ΔI_L = 200mA	V_{OUT}	E_{66}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{66}}{200}$	See figure 16	-	2.5	mV/mA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12: Test conditions V_{IN} (volts)	See figure 12: Test conditions I_L (mA)	See figure 12: Applied test voltages (volts) [Hi-Lo pin potential] 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines: Pins	Value	Unit	Equation	Notes	Limits: Min	Limits: Max	Unit
1 (T_A = 25°C)	V_{OUT1}	1	-8	5	-8	-	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		-5.25	-4.75	V
	V_{OUT2}	2	-8	1000	-8	9.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	-30	5	-30	-	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	-30	100	-30	0.95	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE1}	5	-8	100	-8	0.95	-	-	-			E_5		$V_{RLINE1} = E_5-E_4$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	6	-8	500	-8	4.95	-	-	-			E_6				-	-	
	V_{RLINE2}	7	-25	500	-25	4.95	-	-	-			E_7		$V_{RLINE2} = E_6-E_7$		-50	50	
	V_{RLOAD1}	8	-10	5	-10	-	-	-	-			E_8			See figure 12 waveforms	-	-	
	V_{RLOAD1}	9	-10	1000	-10	9.95	-	-	-			E_9		$V_{RLOAD1} = E_8-E_9$		-100	100	
	V_{RLOAD2}	10	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_3-E_4$		-150	150	
	V_{RTH}	11	-15	1000	-15	9.95	-	-	-		9-11	E_{10}		$V_{RTH} = E_{10}$	See figure 12 waveforms	-50	50	
	I_{SCD1}	12	-10	5	-10	-	-	-	-		12-13	E_{11}		$I_{SCD1} = \frac{E_{11}}{2000}$		0.1	3.0	mA
	I_{SCD2}	13	-30	5	-30	-	-	-	-			E_{12}		$I_{SCD2} = \frac{E_{12}}{2000}$		0.1	4.0	
	ΔI_{SCD} (LINE)	14	-8	5	-8	-	-	-	-			E_{13}		ΔI_{SCD} (LINE) $= \frac{E_{13}-E_{12}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	15	-10	1000	-10	9.95	-	-	-			E_{14}		ΔI_{SCD} (LOAD) $= \frac{E_{11}-E_{14}}{2000}$		-0.5	0.5	
	I_{OS1}	16	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{15}		$I_{OS1} = E_{15}$	See figure 12 waveforms	0.002	4.5	A
	V_{OUT5} (RECOV)	17	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{16}		$V_{OUT5} = E_{16}$		-5.25	-4.75	V
	I_{OS2}	18	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{17}		$I_{OS2} = E_{17}$		0.002	2.0	A
	V_{OUT6} (RECOV)	19	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{18}		$V_{OUT6} = E_{18}$		-5.25	-4.75	V
	I_{CTL}	20	-10	500	-10	4.95	-	-	-	K1,K2	12-13	E_{19}		$I_{CTL} = \frac{E_{19}}{33200}$		0.01	2	µA
	V_{START}	21	-8	1000	-15	-	-	-0.8	-	K4	9-11	E_{20}		$V_{OUT} = E_{20}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	22	-38	1000	-38	-	-	-	-	None		E_{21}		$V_{OUT7} = E_{21}$	R_1 = 5.49k ±0.1%	-31.5	-28.5	
2 (T_A = 125°C)	V_{OUT1}	23	-8	5	-8	-	-	-	-			E_{22}		$V_{OUT1} = E_{22}$		-5.25	-4.75	
	V_{OUT2}	24	-8	1000	-8	9.95	-	-	-			E_{23}		$V_{OUT2} = E_{23}$				
	V_{OUT3}	25	-30	5	-30	-	-	-	-			E_{24}		$V_{OUT3} = E_{24}$				
	V_{OUT4}	26	-30	100	-30	0.95	-	-	-			E_{25}		$V_{OUT4} = E_{25}$				
	V_{RLINE1}	27	-8	100	-8	0.95	-	-	-			E_{26}		$V_{RLINE1} = E_{26}-E_{25}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	28	-8	500	-8	4.95	-	-	-			E_{27}				-	-	
	V_{RLINE2}	29	-25	500	-25	4.95	-	-	-			E_{28}		$V_{RLINE2} = E_{27}-E_{28}$		-50	50	
	V_{RLOAD1}	30	-10	5	-10	-	-	-	-			E_{29}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	31	-10	1000	-10	9.95	-	-	-			E_{30}		$V_{RLOAD1} = E_{29}-E_{30}$		-100	100	
	V_{RLOAD2}	32	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{24}-E_{25}$		-150	150	
	I_{SCD1}	33	-10	5	-10	-					12-13	E_{31}		$I_{SCD1} = \frac{E_{31}}{2000}$		0.1	3.0	mA
	I_{SCD2}	34	-30	5	-30	-	-	-	-		12-13	E_{32}		$I_{SCD2} = \frac{E_{32}}{2000}$		0.1	4.0	mA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
2 (T_A = 125°C)	ΔI_{SCD} (LINE)	35	-8	5	-8	-	-	-	-	None	12-13	E_{33}	V	ΔI_{SCD} (LINE) = $\frac{E_{33}-E_{32}}{2000}$		-1.0	1.0	mA
	ΔI_{SCD} (LOAD)	36	-10	1000	-10	9.95	-	-	-	None	12-13	E_{34}		ΔI_{SCD} (LOAD) = $\frac{E_{31}-E_{34}}{2000}$		-0.5	0.5	mA
	I_{OS1}	37	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{35}		$I_{OS1} = E_{35}$	See figure 12 waveforms	0.002	4.5	A
	V_{OUT5} (RECOV)	38	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{36}		$V_{OUT5} = E_{36}$		-5.25	-4.75	V
	I_{OS2}	39	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{37}		$I_{OS2} = E_{37}$		0.002	2.0	A
	V_{OUT6} (RECOV)	40	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{38}		$V_{OUT6} = E_{38}$		-5.25	-4.75	V
	I_{CTL}	41	-10	500	-10	4.95	-	-	-	K1,K2	12-13	E_{39}		$I_{CTL} = \frac{E_{39}}{33200}$		0.001	3	µA
	V_{START}	42	-8	1000	-15	-	-	-0.8	-	K4	9-11	E_{40}		$V_{OUT} = E_{40}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	43	-38	1000	-38	-	-	-	-	None		E_{41}		$V_{OUT7} = E_{41}$	R_1 = 5.49kΩ ±0.1%	-31.5	-28.5	
(T_A = 150°C)	V_{OUT8}	44	-10	5	-10	-	-	-	-			E_{42}		$V_{OUT8} = E_{42}$		-5.30	-4.70	
3 (T_A = -55°C)	V_{OUT1}	45	-8	5	-8	-	-	-	-			E_{43}		$V_{OUT1} = E_{43}$		-5.25	-4.75	
	V_{OUT2}	46	-8	1000	-8	9.95	-	-	-			E_{44}		$V_{OUT2} = E_{44}$				
	V_{OUT3}	47	-30	5	-30	-	-	-	-			E_{45}		$V_{OUT3} = E_{45}$				
	V_{OUT4}	48	-30	100	-30	0.95	-	-	-			E_{46}		$V_{OUT4} = E_{46}$				
	V_{RLINE1}	49	-8	100	-8	0.95	-	-	-			E_{47}		$V_{RLINE1} = E_{47}-E_{46}$	See figure 12 waveforms	-150	150	mV
	V_{RLINE2}	50	-8	500	-8	4.95	-	-	-			E_{48}				-	-	
	V_{RLINE2}	51	-25	500	-25	4.95	-	-	-			E_{49}		$V_{RLINE2} = E_{48}-E_{49}$		-50	50	
	V_{RLOAD1}	52	-10	5	-10	-	-	-	-			E_{50}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	53	-10	1000	-10	9.95	-	-	-			E_{51}		$V_{RLOAD1} = E_{50}-E_{51}$		-100	100	
	V_{RLOAD2}	54	-	-	-	-	-	-	-		-	-		$V_{RLOAD2} = E_{45}-E_{46}$		-150	150	
	I_{SCD1}	55	-10	5	-10	-	-	-	-		12-13	E_{52}		$I_{SCD1} = \frac{E_{52}}{2000}$		0.1	3.0	mA
	I_{SCD2}	56	-30	5	-30	-	-	-	-			E_{53}		$I_{SCD2} = \frac{E_{53}}{2000}$		0.1	4.0	
	ΔI_{SCD} (LINE)	57	-8	5	-8	-	-	-	-			E_{54}		ΔI_{SCD} (LINE) = $\frac{E_{54}-E_{53}}{2000}$		-1.0	1.0	
	ΔI_{SCD} (LOAD)	58	-10	1000	-10	9.95	-	-	-			E_{55}		ΔI_{SCD} (LOAD) = $\frac{E_{52}-E_{55}}{2000}$		-0.5	0.5	
	I_{OS1}	59	-10	-	-15	-	-	-1.0	0	K4,K5	10-5	E_{56}		$I_{OS1} = E_{56}$	See figure 12 waveforms	0.002	4.5	A
	V_{OUT5} (RECOV)	60	-10	-	-15	-	-	-1.0	0	K4,K5	9-11	E_{57}		$V_{OUT5} = E_{57}$		-5.25	-4.75	V
	I_{OS2}	61	-30	-	-40	-	-	-3.5	0	K5	10-5	E_{58}		$I_{OS2} = E_{58}$		0.002	2.0	A
	V_{OUT6} (RECOV)	62	-30	-	-40	-	-	-3.5	0	K5	9-11	E_{59}		$V_{OUT6} = E_{59}$		-5.25	-4.75	V
	I_{CTL}	63	-10	500	-10	4.95	-	-	-	K1,K2	12-13	E_{60}		$I_{CTL} = \frac{E_{60}}{33200}$		0.001	3	µA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
3 (T_A = -55°C)	V_{START}	64	-8	1000	-15	-	-	-0.8	-	K4	9-11	E_{61}	V	$V_{OUT} = E_{61}$	See figure 12 waveforms	-5.25	-4.75	V
	V_{OUT7}	65	-38	1000	-38	-	-	-	-	None	9-11	E_{62}	V	$V_{OUT7} = E_{62}$	R_1 = 5.49 kΩ ± 0.1%	-31.5	-28.5	V

Subgroup	Symbol	Test no.	Test conditions			Measurement sense lines			Equation	Notes	Limits		Unit
			Input voltage	Load current		Symbol	Value	Unit			Min	Max	
4 (T_A = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = -10V, e_i = 1.0Vrms @ 2400 Hz	I_L = 350mA		e_{orms}	E_{63}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20 \log E_{63}$	See figure 13	45	-	dB
7 (T_A = 25°C)	V_{NO}	67	V_{IN} = -10V	I_L = 100mA		e_{orms}	E_{64}	Vrms	$V_{NO} = E_{64}$	See figure 14	-	250	μVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = -10V, V_{pulse} = -3.0V	I_L = 5mA		V_{OUT}	E_{65}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{65}}{3}$	See figure 15	-	30	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = -10V	I_L = 100mA, ΔI_L = 400mA		V_{OUT}	E_{66}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{66}}{400}$	See figure 16	-	2.5	mV/mA

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
1 (T_A = 25°C)	V_{OUT1}	1	-4.25	5	-4.25	0	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		-1.275	-1.225	V
	V_{OUT2}	2	-4.25	500	-4.25	4.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	-41.25	5	-41.25	0	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	-41.25	50	-41.25	0.45	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE}	5	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 12 waveforms	-9	9	mV
	V_{RLOAD1}	6	-6.25	5	-6.25	0	-	-	-			E_5		———	See figure 12 waveforms	-	-	
	V_{RLOAD1}	7	-6.25	500	-6.25	4.95	-	-	-			E_6		$V_{RLOAD1} = E_5 - E_6$		-12	12	
	V_{RLOAD2}	8	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-6	6	
	V_{RLOAD3}	9	-6.25	200	-6.25	1.95	-	-	-			E_7		$V_{RLOAD3} = E_5 - E_7$		-6	6	
	V_{RTH}	10	-14.6	750	-14.6	7.45	-	-	-			E_8		$V_{RTH} = \Delta_t E_7$	See figure 12 waveforms	-5	5	
	I_{ADJ}	11	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_9	mV	$I_{ADJ} = \frac{E_9}{2000}$		25	100	µA
	I_{ADJ}	12	-41.25	5	-41.25	0	-	-	-			E_{10}		$I_{ADJ} = \frac{E_{10}}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	13	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_9 - E_{10}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	14	-6.25	5	-6.25	0	-	-	-			E_{11}		ΔI_{ADJ} (LOAD) $= \frac{E_{11} - E_{12}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	15	-6.25	500	-6.25	4.95	-	-	-			E_{12}						
	I_{OS1}	16	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{13}	V	$I_{OS1} = E_{13}$	See figure 12 waveforms	0.5	1.8	A
	V_{OUT5} (RECOV)	17	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{14}		V_{OUT5} (RECOV) $= E_{14}$		-1.275	-1.225	V
	I_{OS2}	18	-40	-	-40	-	-	-	0	K5	10-5	E_{15}		$I_{OS2} = E_{15}$		0.05	0.5	A
	V_{OUT6} (RECOV)	19	-40	-	-40	-	-	-	0	K5	9-11	E_{16}		V_{OUT6} (RECOV) $= E_{16}$		-1.275	-1.225	V
	I_{Q1}	20	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{17}		$I_{Q1} = \frac{E_{17}}{2000}$		0.2	3.0	mA
	I_{Q2}	21	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{18}		$I_{Q2} = \frac{E_{18}}{2000}$		0.2	3.0	mA
	I_{Q3}	22	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{19}		$I_{Q3} = \frac{E_{19}}{2000}$		1.0	5.0	mA
	V_{START}	23	-4.25	500	-10	0	-	-0.425	-	K4	9-11	E_{20}		$V_{START} = E_{20}$	See figure 12 waveforms	-1.275	-1.225	V
2 (T_A = 125°C)	V_{OUT1}	24	-4.25	5	-4.25	0	-	-	-	None		E_{21}		$V_{OUT1} = E_{21}$		-1.30	-1.20	
	V_{OUT2}	25	-4.25	500	-4.25	4.95	-	-	-			E_{22}		$V_{OUT2} = E_{22}$				
	V_{OUT3}	26	-41.25	5	-41.25	0	-	-	-			E_{23}		$V_{OUT3} = E_{23}$				
	V_{OUT4}	27	-41.25	50	-41.25	0.45	-	-	-			E_{24}		$V_{OUT4} = E_{24}$				
	V_{RLINE}	28	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{21} - E_{23}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	29	-6.25	5	-6.25	0	-	-	-			E_{25}		———	See figure 12 waveforms	-	-	
	V_{RLOAD1}	30	-6.25	500	-6.25	4.95	-	-	-			E_{26}		$V_{RLOAD1} = E_{25} - E_{26}$		-24	24	
	V_{RLOAD2}	31	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{23} - E_{24}$		-12	12	
	V_{RLOAD3}	32	-6.25	200	-6.25	1.95						E_{27}		$V_{RLOAD3} = E_{25} - E_{27}$		-12	12	

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
2 (T_A = 125°C)	I_{ADJ}	33	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_{28}	mV	$I_{ADJ} = \frac{E_{28}}{2000}$		25	100	µA
	I_{ADJ}	34	-41.25	5	-41.25	0	-	-	-			E_{29}		$I_{ADJ} = \frac{E_{29}}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	35	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{28}-E_{29}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	36	-6.25	5	-6.25	0	-	-	-			E_{30}						
	ΔI_{ADJ} (LOAD)	37	-6.25	500	-6.25	4.95	-	-	-			E_{31}		ΔI_{ADJ} (LOAD) $= \frac{E_{30}-E_{31}}{2000}$		-5	5	
	I_{OS1}	38	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{32}	V	$I_{OS1} = E_{32}$	See figure 12 waveforms	0.5	1.8	A
	V_{OUT5} (RECOV)	39	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{33}		$V_{OUT5} = E_{33}$ (RECOV)		-1.30	-1.20	V
	I_{OS2}	40	-40	-	-40	-	-	-	0	K5	10-5	E_{34}		$I_{OS2} = E_{34}$		0.05	0.5	A
	V_{OUT6} (RECOV)	41	-40	-	-40	-	-	-	0	K5	9-11	E_{35}		$V_{OUT6} = E_{35}$ (RECOV)		-1.30	-1.20	V
	I_{Q1}	42	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{36}		$I_{Q1} = \frac{E_{36}}{2000}$		0.2	3.0	mA
	I_{Q2}	43	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{37}		$I_{Q2} = \frac{E_{37}}{2000}$		0.2	3.0	mA
	I_{Q3}	44	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{38}		$I_{Q3} = \frac{E_{38}}{2000}$		1.0	5.0	mA
	V_{START}	45	-4.25	500	-10	0	-	-0.425	-	K4	9-11	E_{39}		$V_{START} = E_{39}$	See figure 12 waveforms	-1.30	-1.20	V
(T_A = 150°C)	V_{OUT7}	46	-6.25	5	-6.25	0	-	-	-	None		E_{40}		$V_{OUT7} = E_{40}$				
3 (T_A = -55°C)	V_{OUT1}	47	-4.25	5	-4.25	0	-	-	-			E_{41}		$V_{OUT1} = E_{41}$				
	V_{OUT2}	48	-4.25	500	4.25	4.95	-	-	-			E_{42}		$V_{OUT2} = E_{42}$				
	V_{OUT3}	49	-41.25	5	-41.25	0	-	-	-			E_{43}		$V_{OUT3} = E_{43}$				
	V_{OUT4}	50	-41.25	50	-41.25	0.45	-	-	-			E_{44}		$V_{OUT4} = E_{44}$				
	V_{RLINE}	51	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{41}-E_{43}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	52	-6.25	5	-6.25	0	-	-	-			E_{45}			See figure 12 waveforms	-	-	
	V_{RLOAD1}	53	-6.25	500	-6.25	4.95	-	-	-			E_{46}		$V_{RLOAD1} = E_{45}-E_{46}$		-24	24	
	V_{RLOAD2}	54	-	-	-	-	-	-	-			E_{47}		$V_{RLOAD2} = E_{43}-E_{44}$		-12	12	
	V_{RLOAD3}	55	-6.25	200	-6.25	1.95	-	-	-			E_{48}		$V_{RLOAD3} = E_{45}-E_{48}$		-12	12	
	I_{ADJ}	56	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_{49}	mV	$I_{ADJ} = \frac{E_{49}}{2000}$		25	100	µA
	I_{ADJ}	57	-41.25	5	-41.25	0	-	-	-			E_{50}		$I_{ADJ} = \frac{E_{50}}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	58	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{49}-E_{50}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	59	-6.25	5	-6.25	0	-	-	-			E_{51}						
	ΔI_{ADJ} (LOAD)	60	-6.25	500	-6.25	4.95	-	-	-			E_{52}		ΔI_{ADJ} (LOAD) $= \frac{E_{51}-E_{52}}{2000}$		-5	5	

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	See figure 12 Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) [Hi-Lo pin potential] 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines Pins	Value	Unit	Equation	Notes	Limits Min	Max	Unit
3 (T_A = -55°C)	I_{OS1}	61	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{53}	V	$I_{OS1} = E_{53}$	See figure 12 waveforms	0.5	1.8	A
	V_{OUT5} (RECOV)	62	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{54}		$V_{OUT5} = E_{54}$ (RECOV)		-1.30	-1.20	V
	I_{OS2}	63	-40	-	-40	-	-	-	0	K5	10-5	E_{55}		$I_{OS2} = E_{55}$		0.05	0.5	A
	V_{OUT6} (RECOV)	64	-40	-	-40	-	-	-	0	K5	9-11	E_{56}		$V_{OUT6} = E_{56}$ (RECOV)		-1.30	-1.20	V
	I_{Q1}	65	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{57}		$I_{Q1} = \frac{E_{57}}{2000}$		0.2	3.0	mA
	I_{Q2}	66	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{58}		$I_{Q2} = \frac{E_{58}}{2000}$		0.2	3.0	mA
	I_{Q3}	67	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{59}		$I_{Q3} = \frac{E_{59}}{2000}$		1.0	5.0	mA
	V_{START}	68	-4.25	500	-10	0	-	-0.425	-	K4	9-11	E_{60}		$V_{START} = E_{60}$	See figure 12 waveforms	-1.30	-1.20	V

Subgroup	Symbol	Test no.	Test conditions Input voltage	Load current	Measurement sense lines Symbol	Value	Unit	Equation	Notes	Limits Min	Max	Unit
4 (T_A = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	69	V_{IN} = -6.25V, e_i = 1.0Vrms @ 2400 Hz	I_L = 125mA	e_{orms}	E_{61}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20 \log E_{61}$	See figure 13	48	-	dB
7 (T_A = 25°C)	V_{NO}	70	V_{IN} = -6.25V	I_L = 50mA	e_{orms}	E_{62}	Vrms	$V_{NO} = E_{62}$	See figure 14	-	120	µVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	71	V_{IN} = -6.25V, V_{pulse} = -3.0V	I_L = 10mA	V_{OUT}	E_{63}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{63}}{3}$	See figure 15	-	80	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	72	V_{IN} = -6.25V	I_L = 50mA, ΔI_L = 200mA	V_{OUT}	E_{64}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{64}}{200}$	See figure 16	-	0.30	mV/mA

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
1 (T_A = 25°C)	V_{OUT1}	1	-4.25	5	-4.25	0	-	-	-	None	9-11	E_1	V	$V_{OUT1} = E_1$		-1.275	-1.225	V
	V_{OUT2}	2	-4.25	1500	-4.25	14.95	-	-	-			E_2		$V_{OUT2} = E_2$				
	V_{OUT3}	3	-41.25	5	-41.25	0	-	-	-			E_3		$V_{OUT3} = E_3$				
	V_{OUT4}	4	-41.25	200	-41.25	1.95	-	-	-			E_4		$V_{OUT4} = E_4$				
	V_{RLINE}	5	-	-	-	-	-	-	-			-		$V_{RLINE} = E_1 - E_3$	See figure 12 waveforms	-9	9	mV
	V_{RLOAD1}	6	-6.25	5	-6.25	0	-	-	-			E_5		———	See figure 12 waveforms	—	—	
	V_{RLOAD1}	7	-6.25	1500	-6.25	14.95	-	-	-			E_6		$V_{RLOAD1} = E_5 - E_6$		-6	6	
	V_{RLOAD2}	8	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_3 - E_4$		-6	6	
	V_{RTH}	9	-14.6	1500	-14.6	14.95	-	-	-			E_7		$V_{RTH} = E_7$	See figure 12 waveforms	-5	5	
	I_{ADJ}	10	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_8	mV	$I_{ADJ} = \frac{E_8}{2000}$		25	100	µA
	I_{ADJ}	11	-41.25	5	-41.25	0	-	-	-			E_9		$I_{ADJ} = \frac{E_9}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	12	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_8 - E_9}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	13	-6.25	5	-6.25	0	-	-	-			E_{10}						
	ΔI_{ADJ} (LOAD)	14	-6.25	1500	-6.25	14.95	-	-	-			E_{11}		ΔI_{ADJ} (LOAD) $= \frac{E_{10} - E_{11}}{2000}$		-5	5	
	I_{OS1}	15	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{12}	V	$I_{OS} = E_{12}$	See figure 12 waveforms	1.5	3.5	A
	V_{OUT5} (RECOV)	16	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{13}		$V_{OUT5} = E_{13}$ (RECOV)		-1.275	-1.225	V
	I_{OS2}	17	-40	-	-40	-	-	-	0	K5	10-5	E_{14}		$I_{OS} = E_{14}$		0.2	0.8	A
	V_{OUT6} (RECOV)	18	-40	-	-40	-	-	-	0	K5	9-11	E_{15}		$V_{OUT6} = E_{15}$ (RECOV)		-1.275	-1.225	V
	I_{Q1}	19	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{16}		$I_{Q1} = \frac{E_{16}}{2000}$		0.2	3.0	mA
	I_{Q2}	20	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{17}		$I_{Q2} = \frac{E_{17}}{2000}$		0.2	3.0	mA
	I_{Q3}	21	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{18}		$I_{Q3} = \frac{E_{18}}{2000}$		1.0	5.0	mA
	V_{START}	22	-4.25	1500	-10	0	-	-0.425	-	K4	9-11	E_{19}		$V_{START} = E_{19}$	See figure 12 waveforms	-1.275	-1.225	V
2 (T_A = 125°C)	V_{OUT1}	23	-4.25	5	-4.25	0	-	-	-	None		E_{20}		$V_{OUT1} = E_{20}$		-1.30	-1.20	
	V_{OUT2}	24	-4.25	1500	-4.25	14.95	-	-	-			E_{21}		$V_{OUT2} = E_{21}$				
	V_{OUT3}	25	-41.25	5	-41.25	0	-	-	-			E_{22}		$V_{OUT3} = E_{22}$				
	V_{OUT4}	26	-41.25	200	-41.25	1.95	-	-	-			E_{23}		$V_{OUT4} = E_{23}$				
	V_{RLINE}	27	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{20} - E_{22}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	28	-6.25	5	-6.25	0	-	-	-			E_{24}		———	See figure 12 waveforms	—	—	
	V_{RLOAD1}	29	-6.25	1500	-6.25	14.95	-	-	-			E_{25}		$V_{RLOAD1} = E_{24} - E_{25}$		-12	12	
	V_{RLOAD2}	30	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{22} - E_{23}$		-12	12	

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	See figure 12							Relays energized	Measurement sense lines			Equation	Notes	Limits		Unit
			Test conditions		Applied test voltages (volts) [Hi-Lo pin potential]						Pins	Value	Unit			Min	Max	
			V_{IN} (volts)	I_L (mA)	1-2	4-5	6-11	7-2	8-2									
2 (T_A = 125°C)	I_{ADJ}	31	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_{26}	mV	$I_{ADJ} = \frac{E_{26}}{2000}$		25	100	µA
	I_{ADJ}	32	-41.25	5	-41.25	0	-	-	-			E_{27}		$I_{ADJ} = \frac{E_{27}}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	33	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{26}-E_{27}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	34	-6.25	5	-6.25	0	-	-	-			E_{28}		ΔI_{ADJ} (LOAD) $= \frac{E_{28}-E_{29}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	35	-6.25	1500	-6.25	14.95	-	-	-			E_{29}						
	I_{OS1}	36	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{30}	V	$I_{OS1} = E_{30}$	See figure 12 waveforms	1.5	3.5	A
	V_{OUT5} (RECOV)	37	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{31}		$V_{OUT5} = E_{31}$ (RECOV)		-1.30	-1.20	V
	I_{OS2}	38	-40	-	-40	-	-	-	0	K5	10-5	E_{32}		$I_{OS2} = E_{32}$		0.2	0.8	A
	V_{OUT6} (RECOV)	39	-40	-	-40	-	-	-	0	K5	9-11	E_{33}		$V_{OUT6} = E_{33}$ (RECOV)		-1.30	-1.20	V
	I_{Q1}	40	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{34}		$I_{Q1} = \frac{E_{34}}{2000}$		0.2	3.0	mA
	I_{Q2}	41	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{35}		$I_{Q2} = \frac{E_{35}}{2000}$		0.2	3.0	mA
	I_{Q3}	42	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{36}		$I_{Q3} = \frac{E_{36}}{2000}$		1.0	5.0	mA
	V_{START}	43	-4.25	1500	-10	0	-	-0.425	-	K4	9-11	E_{37}		$V_{START} = E_{37}$	See figure 12 waveforms	-1.30	-1.20	V
(T_A = 150°C)	V_{OUT7}	44	-6.25	5	-6.25	0	-	-	-	None		E_{38}		$V_{OUT7} = E_{38}$				
3 (T_A = -55°C)	V_{OUT1}	45	-4.25	5	-4.25	0	-	-	-			E_{39}		$V_{OUT1} = E_{39}$				
	V_{OUT2}	46	-4.25	1500	-4.25	14.95	-	-	-			E_{40}		$V_{OUT2} = E_{40}$				
	V_{OUT3}	47	-41.25	5	-41.25	0	-	-	-			E_{41}		$V_{OUT3} = E_{41}$				
	V_{OUT4}	48	-41.25	200	-41.25	1.95	-	-	-			E_{42}		$V_{OUT4} = E_{42}$				
	V_{RLINE}	49	-	-	-	-	-	-	-			-		$V_{RLINE} = E_{39}-E_{41}$	See figure 12 waveforms	-23	23	mV
	V_{RLOAD1}	50	-6.25	5	-6.25	0	-	-	-			E_{43}		———	See figure 12 waveforms	—	—	
	V_{RLOAD1}	51	-6.25	1500	-6.25	14.95	-	-	-			E_{44}		$V_{RLOAD1} = E_{43}-E_{44}$		-12	12	
	V_{RLOAD2}	52	-	-	-	-	-	-	-			-		$V_{RLOAD2} = E_{41}-E_{42}$		-12	12	
	I_{ADJ}	53	-4.25	5	-4.25	0	-	-	-	K2	12-13	E_{45}	mV	$I_{ADJ} = \frac{E_{45}}{2000}$		25	100	µA
	I_{ADJ}	54	-41.25	5	-41.25	0	-	-	-			E_{46}		$I_{ADJ} = \frac{E_{46}}{2000}$		25	100	
	ΔI_{ADJ} (LINE)	55	-	-	-	-	-	-	-			-		ΔI_{ADJ} (LINE) $= \frac{E_{45}-E_{46}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	56	-6.25	5	-6.25	0	-	-	-			E_{47}		ΔI_{ADJ} (LOAD) $= \frac{E_{47}-E_{48}}{2000}$		-5	5	
	ΔI_{ADJ} (LOAD)	57	-6.25	1500	-6.25	14.95	-	-	-			E_{48}						

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	See figure 12: Test conditions V_{IN} (volts)	I_L (mA)	Applied test voltages (volts) [Hi-Lo pin potential] 1-2	4-5	6-11	7-2	8-2	Relays energized	Measurement sense lines: Pins	Value	Unit	Equation	Notes	Limits: Min	Max	Unit
3 (T_A = -55°C)	I_{OS1}	58	-4.25	-	-10	-	-	-0.425	0	K4,K5	10-5	E_{49}	V	$I_{OS1} = E_{49}$	See figure 12 waveforms	1.5	3.5	A
	V_{OUT5} (RECOV)	59	-4.25	-	-10	-	-	-0.425	0	K4,K5	9-11	E_{50}		$V_{OUT5} = E_{50}$ (RECOV)		-1.30	-1.20	V
	I_{OS2}	60	-40	-	-40	-	-	-	0	K5	10-5	E_{51}		$I_{OS2} = E_{51}$		0.2	0.8	A
	V_{OUT6} (RECOV)	61	-40	-	-40	-	-	-	0	K5	9-11	E_{52}		$V_{OUT6} = E_{52}$ (RECOV)		-1.30	-1.20	V
	I_{Q1}	62	-4.25	-	-4.25	0	-1.4	-	-	K3	12-13	E_{53}		$I_{Q1} = \frac{E_{53}}{2000}$		0.2	3.0	mA
	I_{Q2}	63	-14.25	-	-14.25	0	-1.4	-	-	K3	12-13	E_{54}		$I_{Q2} = \frac{E_{54}}{2000}$		0.2	3.0	mA
	I_{Q3}	64	-41.25	-	-41.25	0	-1.4	-	-	K3	12-13	E_{55}		$I_{Q3} = \frac{E_{55}}{2000}$		1.0	5.0	mA
	V_{START}	65	-4.25	1500	-10	0	-	-0.425	-	K4	9-11	E_{56}		$V_{START} = E_{56}$	See figure 12 waveforms	-1.30	-1.20	V

Subgroup	Symbol	Test no.	Test conditions: Input voltage	Load current	Measurement sense lines: Symbol	Value	Unit	Equation	Notes	Limits: Min	Max	Unit
4 (T_A = 25°C)	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	66	V_{IN} = -6.25V, e_i = 1.0Vrms @ 2400 Hz	I_L = 500mA	e_{orms}	E_{57}	Vrms	$\frac{\Delta V_{IN}}{\Delta V_{OUT}} = -20 \log E_{57}$	See figure 13	50	-	dB
7 (T_A = 25°C)	V_{NO}	67	V_{IN} = -6.25V	I_L = 100mA	e_{orms}	E_{58}	Vrms	$V_{NO} = E_{58}$	See figure 14	-	120	µVrms
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	68	V_{IN} = -6.25V, V_{pulse} = -3.0V	I_L = 10mA	V_{OUT}	E_{59}	V	$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{E_{59}}{3}$	See figure 15	-	40	mV/V
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	69	V_{IN} = -6.25V	I_L = 100mA, ΔI_L = 400mA	V_{OUT}	E_{60}	V	$\frac{\Delta V_{OUT}}{\Delta I_L} = \frac{E_{60}}{200}$	See figure 16	-	0.15	mV/mA

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3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. For class S and B devices, an additional burn-in screen shall be performed in accordance with figure 10 to test the operation of the thermal shutdown circuit. This screen shall be performed after serialization (3.1.8 of method 5004 of MIL-STD-883) and before interim electrical parameters (pre burn-in, 3.1.9 of method 5004 of MIL-STD-883). The requirements of 3.2.3 of method 1015 of MIL-STD-883 shall apply to this screen except the devices need not be tested in an oven.
b. Interim electrical parameters shall be performed for class S and B devices and shall be as specified in table II.
c. Burn-in (method 1015 of MIL-STD-883). For class S and B devices, only test condition C shall apply using the circuit shown on figure 11. Accelerated test conditions shall not be used.
d. Reverse bias burn-in shall not be performed.
e. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA is specified as 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.
f. Constant acceleration (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 5, 6, 8, 9, 10, and 11 shall be omitted.
b. Tests shall be as specified in table II herein.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Life test for class S shall be performed at 125°C for 1000 hours at an LTPD of 5 (per group C, subgroup 2) as a substitute for subgroup 5 of table IIa. If the sample fails the LTPD acceptance, individual devices in the lot may be acceptable if they pass 1000 hours additional burn-in and the lot PDA does not exceed 5 percent.
c. Constant acceleration for class S (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Life tests for class B and C (method 1005 of MIL-STD-883), test condition C only using the circuits shown on figure 11. Accelerated test conditions shall not be used.
c. Constant acceleration (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.4.4 Group D inpection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Constant acceleration (method 2001 of MIL-STD-883); test condition B shall be used for case Y.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the designated return sense line. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

4.6 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with the requirements of MIL-M-38510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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TABLE IV. Group C end-point electrical parameters.

Device type	Characteristic	Symbol	Delta limits 1/	Limits		Unit
				Min	Max	
01,02	Output voltage	V_{OUT}	±50 mV	-5.25	-4.75	V
	Standby current drain	I_{SCD}	±20%	0.5	4.0	mA
03,04	Output voltage	V_{OUT}	±10 mV	-1.30	-1.20	V
	Adjust pin current	I_{ADJ}	±10 µA	25	100	µA
	Line regulation	V_{RLINE}	±4 mV	-9	9	mV

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for packaging and packing.
f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

6.4.1 Line regulation. The change in output voltage for a specified change in input voltage (V_{RLINE}).

6.4.2 Load regulation. The change in output voltage for a specified change in load current (V_{RLOAD}).

6.4.3 Ripple rejection. The ratio of the peak-to-peak input ripple voltage to the peak-to-peak output ripple voltage ($\Delta V_{OUT}/\Delta V_{IN}$).

6.4.4 Output noise voltage. The rms output noise voltage with constant load and no input ripple (V_{NO}).

6.4.5 Standby current drain. The supply current drawn by the regulator with no output load or with a 1-kilohm output load.

6.4.6 Minimum load current. The minimum load current is that current required to maintain regulation.

6.4.7 Input voltage range. The range of supply voltage over which the regulator will operate.

6.4.8 Output voltage range. The range of output voltage over which the regulator will operate.

6.4.9 Transient response. The closed-loop step function response of the regulator under small-signal conditions.

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	79MG
02	79G
03	LM137H
04	LM137K

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI, AR
Navy - SH
Air Force - 99
DLA - ES
NASA - NS

User activities:
Army - SM
Navy - CG, MC, AS, OS
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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